Main Group Work_ML

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Next Steps:

Pascal: - – a Generalised Linear Model with family set to Poisson –> glm(Price ~ reviews) Achtung Price kann nicht mit Poisson gerechnet werden! Sind keine Zähldaten. Neu Auslastung genutzt, um Zähldaten zu haben.

Timon: - a Linear Model –> lm(price ~ rewview_rating + Superhost T/F) –> für unterscheidung und kontrolle.

Davide: - – a Generalised Linear Model with family set to Binomial –> glm(Price ~ Superhost T/F)

Modelle zu verteilen: – a Generalised Additive Model –> gam(price ~ s(accommodates) + –> allenfalls aufbauend auf dem lm() s(review_scores_rating) + s(availability_365) +
s(number_of_reviews) +
s(longitude, latitude) +
s(host_listings_count) +
room_type +
host_is_superhost +
bathrooms + bedrooms,
family = gaussian(link="log"),
data = df_clean)

– a Neural Network –> Ev. auch Kategorisierung analog SVM – a Support Vector Machine –> Overpriced & Underpriced & Right Price (Davide und Timon)

Infos zu bedenken: ABGABEDATUM: 12 Juni

- Outlier Handling definieren. –> falsche Datenpunkte sollten gelöscht werden. Extremwerte im Zweifel drin lassen. Ansonsten kann das als p-hacking verstanden werden.

Coding Conventions. • data sets are named with the prefix "d." (e.g., d.cats). • linear model objects are named with the prefix "lm." (e.g., lm.cats) • within the text R-objects are written using code formatting (e.g., lm.cats) 14 • variable names are written using italic (e.g., Body.weight) • these conventions are arbitrary, but hopefully useful to the reader

#Research Questions

What factors, which can be modified by Hosts on AirBnb, affect guest willingness to pay more for a certain room?

What are factors that influence the guest willingness to pay (WTP)

What are factors on which the guest can adjust their pricing model?

##Concepts: Willingness to pay (WTP)

WTP is the hypothetical value, how much a customer is willing to pay for a location or a room for a certain standard. This can rise, if for example the host provides the ability to wash clothes in the apartment. So for the same location, once with and once without a washing machine, the customers WTP might increase with

the washer. In a more abstract example, if the host is a super host, the customer might trust the host more and his WTP is higher.

##Hypothesis H1: If the review-score of a host is better, the WTP of a guest increases. H2: If a host has more reviews, the WTP of a guest increases. H3: If a host is e superhost, the WTP of a guest increases. H4:

WTP is measured in the Price displayed. We assume, that the mean of prices is similar to the WTP and residuals even out. Therefore we can work with the effective prices displayed on AirBnb.

Business Case

We propose a data-driven pricing decision-support system for Airbnb hosts in the Canton of Vaud, providing market-aligned price recommendations based on listing characteristics. The tool targets both new and existing hosts, using inputs such as location, room type, guest capacity, availability, ratings, and superhost status. Machine learning models, including LM, GLM, GAM, SVM, and Neural Networks, will be applied to identify key price drivers and generate accurate predictions. The objective is to balance revenue and occupancy, helping hosts understand their market value. A key feature is interpretability, enabling users to see why a certain price is recommended. From a business perspective, the system aims to increase booking rates, reduce price dispersion, and improve host satisfaction. The model is tailored to the Canton of Vaud to capture regional economic effects.

First step: Loading Packages

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(janitor)
```

```
Attache Paket: 'janitor'

Die folgenden Objekte sind maskiert von 'package:stats':

    chisq.test, fisher.test
```

```r
library(mgcv)
```

```
Lade nötiges Paket: nlme

Attache Paket: 'nlme'

Das folgende Objekt ist maskiert 'package:dplyr':

    collapse

This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```r
library(e1071)
```

```
Attache Paket: 'e1071'

Das folgende Objekt ist maskiert 'package:ggplot2':

    element
```

```r
library(nnet)
```

```
Attache Paket: 'nnet'

Das folgende Objekt ist maskiert 'package:mgcv':

    multinom
```

```r
library(pROC)
```

```
Type 'citation("pROC")' for a citation.

Attache Paket: 'pROC'

Die folgenden Objekte sind maskiert von 'package:stats':

    cov, smooth, var
```

```r
library(Metrics)
```

```
Attache Paket: 'Metrics'

Das folgende Objekt ist maskiert 'package:pROC':

    auc
```

```r
library(performance)
```

```
Attache Paket: 'performance'

Die folgenden Objekte sind maskiert von 'package:Metrics':

    mae, mse, rmse
```

```r
library(broom)
library(stringr)
library(ggplot2)
library(patchwork)
library(knitr)
```

Second step: Reading all available Datasets

```r
path_listings <- "./Data_Vaud_all/listings-2.csv"
d.raw_data_listings <- read.csv(path_listings, header = TRUE)
```

Cleaning the Airbnb-Listings Data.

```r
d.airbnb_listings <- d.raw_data_listings %>%
  clean_names()

d.listings_clean <- d.airbnb_listings %>%
  mutate(
    price = str_remove_all(price, "\\$|,"),
    price = as.numeric(price),
    host_is_superhost = ifelse(host_is_superhost %in% c("t", "TRUE", "1"), 1, 0),
    room_type = as.factor(room_type),
    property_type = as.factor(property_type)
  ) %>%
  filter(!is.na(price), price > 0)

d.listings_clean <- d.listings_clean %>%
  select(-id, -host_id, -host_listings_count, -host_total_listings_count, -listing_url, -scrape_id, -las
         -source, -name, -description, -maximum_nights, -first_review, -last_review, -calculated_host_li
         -neighborhood_overview, -picture_url, -host_url, -host_name, -host_location,
         -host_about, -host_thumbnail_url, -host_picture_url, -host_neighbourhood,
         -neighbourhood, -neighbourhood_cleansed, -bathrooms_text, -amenities,
         -minimum_minimum_nights, -maximum_minimum_nights, -minimum_maximum_nights,
         -maximum_maximum_nights, -minimum_nights_avg_ntm, -maximum_nights_avg_ntm,
         -calendar_updated, -has_availability, -availability_30, -availability_60,
         -availability_90, -calendar_last_scraped, -number_of_reviews_ltm, -number_of_reviews_l30d,
         -availability_eoy, -number_of_reviews_ly, -review_scores_accuracy, -review_scores_cleanliness,
         -review_scores_checkin, -review_scores_communication, -review_scores_location,
         -license, -instant_bookable, -calculated_host_listings_count_entire_homes, -calculated_host_lis
         -calculated_host_listings_count_shared_rooms, -reviews_per_month, -bedrooms, -review_scores_val
```

Das ist unser definitives Datenset!

```r
str(d.listings_clean)
```

```
'data.frame':   3876 obs. of  20 variables:
 $ host_response_time          : chr  "within an hour" "N/A" "within a few hours" "N/A" ...
 $ host_response_rate          : chr  "100%" "N/A" "100%" "N/A" ...
 $ host_acceptance_rate        : chr  "100%" "N/A" "79%" "100%" ...
 $ host_is_superhost           : num  0 0 1 0 1 1 1 1 1 1 ...
 $ host_verifications          : chr  "['email', 'phone']" "['email', 'phone']" "['email', 'phone']" "[
 $ host_has_profile_pic        : chr  "t" "t" "t" "t" ...
 $ host_identity_verified      : chr  "t" "t" "t" "t" ...
 $ neighbourhood_group_cleansed: chr  "Aigle" "Nyon" "Aigle" "Lausanne" ...
 $ latitude                    : num  46.4 46.5 46.3 46.5 46.5 ...
 $ longitude                   : num  7.13 6.33 7.06 6.64 6.8 ...
 $ property_type               : Factor w/ 63 levels "Barn","Boat",..: 18 44 18 16 52 18 18 44 11 18 ..
 $ room_type                   : Factor w/ 4 levels "Entire home/apt",..: 1 3 1 1 2 1 1 3 1 1 ...
 $ accommodates                : int  3 5 6 3 4 2 2 1 3 14 ...
 $ bathrooms                   : num  1 NA 3 1.5 1 1 1 1 1 2.5 ...
 $ beds                        : int  2 4 4 1 3 2 1 1 2 10 ...
 $ price                       : num  107 725 590 144 254 277 101 34 127 278 ...
 $ minimum_nights              : int  3 4 4 3 1 1 5 2 2 2 ...
 $ number_of_reviews           : int  70 0 18 22 85 125 26 292 41 197 ...
 $ estimated_occupancy_l365d   : int  42 0 0 0 42 120 0 78 150 78 ...
 $ review_scores_rating        : num  4.74 NA 4.94 4.68 4.75 4.87 4.23 4.67 4.95 4.68 ...
```

First Feature Engineering for the airbnb_listings data

```r
#Für Model XXXX
d.airbnb_listings_feat <- d.listings_clean %>%
  mutate(
    log_price = log(price),
    price_per_guest = price / pmax(accommodates, 1),
    high_demand = ifelse(estimated_occupancy_l365d > 200, 1, 0)
  )

#Für LM (Timon)

#Umbauen der Variablen von Character-Prozent in binominale Werte dieser beider
#allenfalls in die DAtenaufbereitung verschieben?????
d.airbnb_listings_feat$host_response_rate <- as.numeric(sub("%", "", d.airbnb_listings_feat$host_respons
```

```
Warning: NAs durch Umwandlung erzeugt
```

```r
print(d.airbnb_listings_feat$host_response_rate)
```

```
   [1] 1.00   NA 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.90 1.00 0.90 0.90
  [15] 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA   NA 1.00 0.00 1.00 0.50   NA
  [29] 0.80 1.00 1.00 0.00 1.00 1.00 1.00 1.00 1.00 1.00 0.90   NA 1.00 1.00
  [43] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.90 1.00 1.00
  [57] 1.00 1.00 0.00 1.00 1.00   NA 1.00 0.50   NA 1.00   NA 1.00 1.00   NA
  [71] 1.00 1.00   NA 1.00   NA 1.00 1.00   NA   NA 0.80 1.00 1.00 1.00   NA
  [85] 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00
  [99] 1.00   NA 0.96 0.96 1.00 1.00 0.90 1.00 1.00 0.90   NA 1.00 1.00 1.00
 [113]   NA 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [127] 0.00   NA 1.00 1.00 1.00   NA 1.00 1.00   NA   NA 1.00 1.00 1.00   NA
 [141] 0.90 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00   NA 1.00 0.96 1.00 1.00
 [155] 1.00   NA   NA 1.00 1.00 0.90 0.80 1.00   NA   NA 1.00 1.00 1.00 0.80
 [169] 0.80 0.80 0.80 1.00 1.00 1.00 1.00 1.00 1.00 0.80 1.00 1.00 1.00 1.00
 [183] 1.00 1.00 1.00 1.00 1.00 0.00 0.67 1.00   NA 1.00 0.96   NA 1.00 1.00
 [197] 0.90 1.00 1.00   NA 1.00 0.50   NA 1.00 1.00 1.00   NA 0.67 1.00 1.00
 [211] 0.90 1.00 1.00 0.80   NA 1.00   NA 0.90 1.00 1.00 1.00 1.00 0.60 1.00
 [225]   NA 1.00   NA 0.50 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00
 [239]   NA 1.00   NA 1.00   NA   NA 1.00 1.00 1.00 1.00   NA 0.00 0.90 0.70
 [253]   NA 0.90 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [267] 1.00 0.33 1.00 1.00 0.70 1.00 1.00   NA 0.50 1.00   NA 0.00 1.00   NA
 [281] 1.00 1.00 0.00 0.50 1.00 1.00 1.00   NA 0.70 1.00   NA 1.00 1.00 1.00
 [295] 0.80 1.00 0.90 0.86 1.00   NA 0.90   NA 0.83 1.00 1.00 1.00 1.00 1.00
 [309] 1.00 1.00 0.96   NA   NA   NA 1.00 1.00 1.00 1.00 1.00 0.80 1.00 1.00
 [323] 1.00 0.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00   NA 0.80
 [337] 0.80 0.80 1.00   NA 0.90 1.00   NA 0.67 1.00 0.90 1.00   NA   NA 1.00
 [351] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.96 1.00 0.50 0.00 1.00 1.00 1.00
 [365] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [379] 1.00 1.00 1.00 1.00   NA 0.96 1.00 1.00 1.00 1.00 0.90 0.00 1.00 0.71
 [393] 0.67   NA   NA 0.00 0.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00
 [407] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 1.00
 [421] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [435] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [449] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00
 [463] 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [477] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA   NA 1.00   NA
 [491] 1.00   NA 1.00 1.00 1.00 1.00 0.90 1.00   NA 1.00 1.00 1.00 1.00 0.00
```

```
 [505] 1.00 1.00 1.00 1.00 1.00 1.00 0.80 1.00 0.75 1.00 1.00   NA 1.00 1.00
 [519] 1.00 0.98   NA 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.75 0.75 0.75
 [533] 0.75 1.00   NA 0.90 0.90 0.50 1.00 0.50 0.00 1.00 0.75 1.00 0.40 1.00
 [547] 1.00 1.00 1.00   NA 1.00 0.80 1.00 1.00 0.98 0.67 0.83 1.00   NA 1.00
 [561] 1.00 1.00   NA 1.00 1.00 1.00 1.00 0.50 1.00 1.00   NA 1.00 1.00 1.00
 [575] 0.98 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA
 [589] 1.00 1.00 0.50 1.00 1.00 1.00 0.88 1.00 1.00 1.00   NA   NA 1.00 1.00
 [603] 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [617] 1.00 1.00 1.00 1.00   NA 1.00   NA   NA 0.97 1.00   NA 1.00 1.00 0.96
 [631] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.75 1.00 1.00 1.00 1.00   NA 1.00
 [645] 1.00 1.00 1.00   NA 1.00 0.80   NA 0.50 1.00   NA   NA   NA 1.00 1.00
 [659] 1.00 0.95 1.00 0.10 1.00 0.50 1.00 1.00 0.96 1.00 1.00 1.00 1.00 1.00
 [673] 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00   NA   NA 1.00   NA 1.00 1.00
 [687] 0.70 1.00 0.50 1.00 1.00 0.90 0.90 0.90 1.00   NA 0.96   NA 1.00 0.98
 [701] 1.00 1.00 1.00 1.00 1.00 0.00 1.00 1.00 1.00 0.00 1.00   NA 1.00 0.80
 [715]   NA   NA 1.00 1.00 1.00 0.88 1.00 1.00 1.00 1.00 0.90 0.90 0.90 0.90
 [729] 1.00 1.00 0.97 1.00 1.00   NA 1.00 1.00 0.96 0.50 1.00   NA   NA 1.00
 [743] 1.00   NA 0.50 1.00 1.00   NA 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.90
 [757] 0.89 1.00 1.00   NA 1.00 0.98 0.98 0.00 1.00 1.00   NA   NA 1.00 0.86
 [771] 1.00 1.00 0.98 1.00 1.00   NA 1.00 0.85 1.00 0.50 1.00 1.00 1.00   NA
 [785]   NA 1.00   NA 0.83 1.00 1.00   NA 1.00 1.00 1.00   NA 1.00 1.00 1.00
 [799] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.00 0.75 1.00 1.00 0.94 1.00   NA
 [813] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.95 1.00 1.00 1.00   NA 1.00
 [827] 1.00   NA 0.50   NA 1.00 1.00 1.00 0.00 1.00 1.00 1.00   NA 1.00 0.00
 [841] 1.00 0.80 1.00 1.00 1.00 0.57 0.73 1.00 1.00 1.00 1.00 1.00 1.00 1.00
 [855]   NA 1.00 0.99   NA 1.00 1.00 1.00 0.00 1.00 1.00 1.00 1.00 0.50   NA
 [869]   NA   NA 1.00   NA 1.00 0.67   NA 0.98 1.00 1.00   NA 1.00 1.00 0.90
 [883] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.75 0.00 1.00 1.00 1.00 1.00
 [897] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.29 1.00   NA   NA 1.00 1.00
 [911] 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 0.93 1.00 1.00 1.00 1.00 1.00
 [925] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.50
 [939] 1.00 1.00 1.00 1.00 1.00 0.96 0.71 1.00 1.00 1.00   NA 1.00 0.75 1.00
 [953] 1.00 1.00 0.97   NA 1.00   NA 1.00 0.00 1.00 0.95 1.00 1.00 1.00 1.00
 [967] 0.80 1.00   NA   NA 1.00 1.00 1.00 1.00 1.00 0.83 1.00 0.75 0.93 1.00
 [981] 1.00   NA 1.00 1.00   NA 0.75 0.90 1.00 1.00 1.00 1.00 0.75 1.00 0.50
 [995]   NA 1.00 0.75 0.75 1.00 0.90 0.90 0.90 0.90 0.90 0.90 0.90 0.90 0.90
[1009] 0.90 1.00 1.00 1.00 0.98 1.00 1.00 1.00 1.00 0.43 0.96 1.00 0.50 1.00
[1023] 1.00 1.00 0.90 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 1.00 0.94 0.94
[1037] 1.00 1.00 1.00 1.00 1.00 1.00 0.83 1.00 1.00 1.00 0.80 0.80 0.80 1.00
[1051] 1.00   NA   NA 1.00   NA   NA 1.00 1.00 1.00 1.00   NA 1.00 1.00   NA
[1065] 1.00 1.00 0.97 0.97 0.97 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00   NA
[1079] 1.00 0.00 1.00   NA 1.00 1.00   NA   NA 1.00 1.00 0.83 1.00 1.00 1.00
[1093] 0.50 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1107]   NA 1.00 1.00 1.00 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1121] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.93 0.93 1.00 1.00 1.00
[1135] 1.00 0.00 0.75 0.98   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1149] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.71
[1163] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00
[1177] 1.00 1.00 1.00   NA   NA 0.98   NA 1.00 1.00 1.00   NA   NA 0.90 1.00
[1191] 1.00 0.50 1.00 1.00 1.00 1.00 0.99 1.00   NA 1.00 0.50 0.98 1.00   NA
[1205] 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00   NA 1.00 0.93 1.00 1.00 1.00
[1219] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.97 1.00 1.00 1.00 1.00 1.00 1.00
[1233] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1247] 0.93 0.00   NA 0.67 0.00 1.00 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
```

```
[1261] 0.75 1.00 1.00 0.10 1.00 0.73 1.00   NA 0.50   NA   NA 0.90 0.90 1.00
[1275] 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.90 1.00
[1289] 0.90 1.00   NA 1.00 0.98 0.98 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00
[1303]   NA   NA 0.60 1.00 0.00   NA   NA   NA 1.00 1.00 1.00 1.00 1.00 1.00
[1317] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1331] 1.00   NA 1.00 1.00 1.00 0.98 1.00 1.00   NA 1.00 1.00 1.00 1.00 0.10
[1345] 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1359]   NA 0.88 1.00 1.00 1.00 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88   NA
[1373] 1.00 1.00 1.00 1.00 1.00 0.90 1.00 1.00 0.67 0.88 1.00 1.00   NA 1.00
[1387] 1.00   NA 1.00 1.00 0.00 0.90   NA 0.86   NA   NA 1.00   NA 1.00 1.00
[1401] 0.00 0.67   NA 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.77 1.00 0.75
[1415] 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 0.80 1.00 1.00 1.00
[1429] 1.00 1.00 0.00 1.00 1.00   NA 1.00 1.00 1.00 0.00 1.00 1.00 1.00 1.00
[1443] 1.00 0.75   NA 1.00 1.00 0.00   NA 1.00 1.00 0.89 1.00 1.00 1.00 1.00
[1457] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.74 1.00 1.00 1.00
[1471] 1.00 1.00 0.90 1.00 1.00 1.00 1.00 1.00 1.00   NA 0.00 1.00 1.00 0.83
[1485] 1.00   NA 0.93 0.93 1.00 0.90 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00
[1499] 1.00 1.00 1.00 1.00 0.90 1.00 1.00 1.00 1.00 0.93 0.00 0.50 0.50 1.00
[1513]   NA 1.00 1.00 1.00 1.00 1.00 0.67   NA 1.00 1.00 0.67 1.00 1.00 1.00
[1527] 0.98 1.00 1.00   NA 1.00 1.00   NA 1.00   NA 1.00 0.98 0.98 0.98 0.90
[1541] 0.00 1.00 1.00 1.00 0.70 1.00 0.80 0.80 0.80 0.70 1.00 1.00 1.00 1.00
[1555] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.70 1.00 1.00
[1569] 1.00 1.00 0.89 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.93 1.00 0.93
[1583] 1.00   NA 1.00 0.93 0.93 0.93 1.00 0.00 1.00 1.00 0.77 1.00   NA 0.00
[1597] 1.00 1.00 1.00 1.00 0.00   NA 0.00   NA 0.80 1.00 1.00 1.00 0.00   NA
[1611]   NA 0.90 0.93   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA   NA
[1625] 1.00   NA 0.50 1.00 1.00 0.80 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.77
[1639]   NA 1.00 1.00   NA   NA 0.00 0.00 0.67 1.00 0.75 0.75 1.00 1.00 0.98
[1653] 0.98 0.98 0.98 0.98 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1667] 1.00 0.00 1.00 1.00   NA 1.00 1.00 1.00 0.50 1.00 1.00 0.78   NA 1.00
[1681] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.80 1.00 0.40 1.00 0.75 1.00 1.00
[1695] 1.00 1.00 1.00 1.00 0.93 0.93 0.93 0.93 0.93 0.93 1.00   NA 1.00 1.00
[1709] 1.00 1.00 0.60 1.00 1.00 1.00 0.67 1.00 1.00 1.00 1.00 0.10 1.00 0.67
[1723] 1.00 0.00 1.00   NA 1.00   NA 1.00   NA 1.00 1.00 0.90 0.70 0.00 1.00
[1737] 1.00 1.00 0.00 1.00   NA 1.00 0.90 1.00 0.00 1.00 0.80   NA 1.00 1.00
[1751] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.60   NA 0.94 1.00   NA 1.00
[1765] 1.00 0.00 1.00 1.00 1.00 1.00 0.96 0.96 0.96 0.43 0.86 0.00 1.00 1.00
[1779] 1.00 0.70 0.00 1.00 1.00 1.00 1.00 0.00 1.00 1.00 1.00 1.00   NA 1.00
[1793] 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.71 0.10 1.00 1.00 0.50 0.98
[1807] 1.00 1.00 0.17 1.00   NA 1.00 1.00 1.00 0.80 1.00 1.00 1.00 1.00 0.90
[1821] 1.00 1.00 1.00 1.00 0.00 1.00 1.00 1.00   NA   NA 1.00 0.96 1.00 1.00
[1835] 1.00 1.00 1.00 1.00 1.00 0.90 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00
[1849] 1.00 1.00 1.00 0.10 1.00 1.00 1.00 1.00 0.00 1.00 1.00 0.00 1.00 1.00
[1863] 1.00 1.00 1.00 1.00 1.00 1.00 0.70   NA 0.70 1.00 1.00 0.90 0.88   NA
[1877] 1.00 0.96 0.96   NA 1.00   NA 1.00 0.90 0.10 1.00 1.00 1.00 1.00 1.00
[1891] 1.00 1.00 1.00 0.90 0.90 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1905]   NA 1.00 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 1.00 1.00 0.40 1.00
[1919] 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 0.86 1.00
[1933] 1.00 1.00 1.00 1.00 0.98   NA 1.00 0.67 0.67 1.00 1.00 0.00 1.00   NA
[1947]   NA 0.96 0.96 0.96 0.67 1.00 1.00   NA   NA 1.00 1.00 1.00 0.90 0.70
[1961] 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00   NA 1.00 1.00   NA
[1975] 1.00 1.00 0.96 1.00 0.96 0.00 1.00 1.00 0.90 1.00 1.00 1.00 1.00 0.00
[1989] 0.70 1.00 1.00 1.00 1.00 1.00 1.00   NA   NA 1.00 0.00 1.00 1.00 0.00
[2003]   NA 0.90 1.00 0.90   NA 1.00 1.00   NA 1.00 1.00 1.00 0.50 1.00 0.33
```

```
[2017] 1.00 0.90 1.00 1.00   NA   NA 1.00   NA   NA   NA 0.00 1.00 1.00 1.00
[2031] 0.90 1.00 1.00 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 1.00 1.00 0.00
[2045]   NA 0.00 1.00 1.00 0.50 1.00 1.00 0.90 0.00 0.80 1.00 0.00 1.00 1.00
[2059] 1.00 1.00 1.00 1.00 1.00 0.33 0.80 1.00 1.00 1.00 1.00 1.00   NA 1.00
[2073] 1.00 1.00   NA 1.00   NA 1.00 0.00 1.00 0.00 0.90 1.00 0.00 1.00 1.00
[2087] 0.10 0.91 0.00 0.90   NA 1.00 1.00 0.17 0.99 1.00   NA 1.00 1.00 0.00
[2101] 0.75 1.00 0.90 1.00 1.00 0.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00
[2115] 1.00 1.00 0.00 1.00 1.00 1.00 0.10 1.00 0.75 1.00 1.00 1.00 1.00 0.00
[2129] 0.80 0.80   NA 0.00   NA 1.00 0.25 1.00 1.00 0.00 0.75 0.83 1.00 0.80
[2143] 0.00 1.00 1.00 1.00 0.93 1.00 1.00 0.99 0.90 0.00   NA 0.75 1.00 0.93
[2157] 0.67 0.98 0.40 1.00 1.00 0.00 0.90   NA 0.00 0.96 1.00 1.00 1.00 0.96
[2171] 1.00 1.00 0.96 1.00 1.00 0.67 0.67   NA 1.00 1.00 0.00 0.75 1.00   NA
[2185] 1.00 1.00 1.00 1.00 1.00 0.93 1.00 0.75 0.00 0.00 1.00 0.00 1.00 0.93
[2199] 0.67 1.00 1.00 0.00 1.00 1.00 0.00 0.50 0.92 1.00 1.00 1.00 1.00 1.00
[2213] 0.93 0.40 0.71 0.60 0.50 1.00 1.00 1.00 1.00 0.93   NA 1.00 0.93 0.93
[2227] 0.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA
[2241] 1.00   NA 1.00 1.00 1.00 1.00 0.13 0.13 1.00   NA 0.70 0.96 1.00 1.00
[2255]   NA   NA 1.00 0.96 1.00 1.00 1.00 1.00 1.00 0.86 1.00 1.00   NA 1.00
[2269] 1.00 1.00 1.00 0.17 0.17   NA 0.10 0.33 1.00 1.00   NA 0.60 1.00 0.43
[2283] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00
[2297] 0.00   NA 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[2311] 1.00 1.00 1.00 1.00 1.00 1.00   NA 0.93 1.00   NA   NA 1.00 1.00 1.00
[2325] 1.00   NA 0.92 0.92 1.00 1.00 1.00 0.60 1.00 1.00 0.00   NA 1.00 1.00
[2339] 1.00   NA 1.00 0.90 0.75 0.75 0.91 1.00 1.00   NA 1.00 1.00 1.00 0.00
[2353]   NA 1.00 1.00 1.00 1.00 0.80 1.00 1.00 1.00 0.78 0.90 1.00 1.00 1.00
[2367]   NA 1.00 1.00 0.90 1.00 0.93 0.93   NA   NA 1.00 1.00 1.00 0.90 0.90
[2381] 0.90 1.00 1.00 0.93 1.00 0.90   NA 1.00 1.00 1.00 1.00 1.00 0.50 1.00
[2395] 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.80 0.67 1.00 0.00
[2409] 1.00 1.00 0.96 0.67   NA 1.00 1.00 1.00 0.93 1.00 0.90 1.00 0.50 1.00
[2423] 0.90 0.98 0.93 1.00 0.93 0.93 0.93 1.00 0.93 0.93 0.93 0.93 0.93 0.93
[2437] 0.93 0.93 0.93 0.93 0.93 0.90 0.00 1.00 1.00 0.98 0.90 1.00 1.00   NA
[2451] 1.00 1.00 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93
[2465] 1.00   NA 1.00 1.00 1.00   NA 0.93 0.96 0.93 0.93 0.93 0.93 1.00 1.00
[2479]   NA   NA   NA 1.00 1.00 0.00 1.00 0.00 0.50   NA 1.00 1.00 0.50 1.00
[2493]   NA 0.90 1.00 1.00 1.00 1.00 0.50 1.00   NA 0.70 0.70 0.70 1.00 1.00
[2507] 1.00 1.00 1.00   NA 1.00 1.00   NA 1.00 1.00   NA 1.00 1.00 1.00 1.00
[2521] 0.50 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 1.00 0.70 0.70
[2535] 1.00 1.00 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88 0.88
[2549] 0.88 0.88 0.88 0.88   NA 1.00 1.00   NA 1.00 1.00 0.00 1.00 0.90 1.00
[2563] 1.00   NA 1.00 1.00 1.00 1.00 1.00   NA 1.00 0.70 1.00 1.00 0.98 1.00
[2577] 1.00 1.00 1.00 0.96 0.96 1.00 1.00 0.46 1.00   NA 0.10 1.00 1.00   NA
[2591] 1.00 1.00 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 1.00 1.00 0.00 1.00
[2605] 1.00 0.00 1.00 1.00   NA   NA 1.00 0.33 1.00 0.50 1.00 1.00   NA 1.00
[2619] 1.00 0.33 1.00 0.00 1.00 0.00 0.80 1.00 0.00 1.00 1.00   NA 1.00   NA
[2633] 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 0.50 1.00 1.00 1.00 0.33   NA
[2647]   NA 0.73 1.00 1.00 0.00 1.00 1.00 1.00 1.00 1.00 1.00 0.00   NA 1.00
[2661] 1.00 0.90 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.00
[2675] 0.25 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00   NA 0.86 1.00
[2689]   NA 1.00 1.00 1.00   NA 0.98 1.00   NA 1.00 1.00 0.00 0.90 0.90 0.96
[2703] 1.00 1.00 1.00 0.00   NA 1.00 1.00 0.20 1.00   NA   NA 0.96   NA 0.50
[2717] 1.00 0.00 0.96   NA 1.00 1.00 1.00 1.00 1.00 1.00   NA 0.80 1.00 0.10
[2731] 0.90 1.00 1.00 1.00 1.00 0.20 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[2745] 1.00 1.00 0.90 0.50 1.00 0.95 0.50 1.00 1.00 0.00 1.00 1.00 0.95 1.00
[2759] 1.00 0.40 1.00 1.00 1.00 1.00 1.00 0.90 0.33 0.50 0.43 0.50 1.00 1.00
```

```
[2773] 0.86 0.96 1.00 1.00 0.97 1.00 0.90 1.00 1.00 1.00 0.96 0.90 0.80 1.00
[2787] 1.00   NA 0.96 0.96 0.96 0.96 0.96 0.96 0.96 1.00 1.00 1.00 1.00 0.00
[2801] 0.50 0.00   NA 0.50 1.00 0.80 1.00 1.00 0.10 1.00 0.86 1.00 1.00   NA
[2815] 1.00 1.00 1.00 1.00 1.00   NA 1.00 0.86 1.00 1.00 1.00   NA 0.00 1.00
[2829] 0.00 1.00 0.90 0.00 1.00 1.00 1.00 1.00   NA 1.00   NA   NA 0.90 0.00
[2843] 0.75 1.00 1.00 1.00   NA   NA 0.82   NA 0.96 1.00 0.70 0.00 0.00 0.93
[2857] 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93
[2871] 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.93 0.00   NA 1.00 1.00
[2885] 1.00 0.96 1.00 1.00 1.00 0.96 1.00 0.96 0.96 1.00 1.00 0.00 1.00 1.00
[2899] 1.00 0.00 1.00 0.00   NA 1.00 1.00 1.00 1.00 1.00 0.00 0.00 1.00 1.00
[2913] 1.00 1.00 1.00 1.00 1.00 0.50 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA
[2927] 1.00   NA 1.00   NA 1.00 1.00 1.00 0.90 1.00 1.00 1.00 0.38   NA 1.00
[2941] 0.50 1.00   NA 1.00 1.00 1.00 0.80 0.00 1.00 1.00 1.00 1.00 1.00 0.99
[2955] 1.00 1.00 1.00 1.00 1.00 0.96 1.00 0.96 0.96 1.00   NA 1.00 1.00 1.00
[2969] 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.50 0.90 1.00 1.00
[2983] 1.00 1.00 1.00   NA   NA 0.30   NA 0.90 0.90 1.00 1.00 0.83 0.90 0.50
[2997] 1.00 1.00 1.00 1.00 0.80 1.00 0.96 0.90 1.00 1.00 1.00 1.00 1.00 0.60
[3011] 0.96 1.00 0.70 0.00 1.00 1.00 1.00 0.95 0.96 1.00 1.00 1.00   NA 1.00
[3025] 0.95 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 0.00 1.00 0.38   NA 0.90
[3039] 0.88 0.88 0.88 1.00 1.00 0.80 1.00 1.00   NA 1.00 1.00 0.00 1.00 1.00
[3053] 1.00 1.00 1.00 0.90 0.96 0.96 0.00 0.67 1.00 1.00 1.00 0.90 1.00 0.90
[3067] 1.00 1.00 0.96 1.00 1.00 1.00 0.00 1.00 0.00 1.00 1.00 0.80 1.00 0.96
[3081] 0.96 1.00 1.00 1.00 0.67 0.00 1.00 1.00 0.80 1.00 1.00 0.78 0.96 1.00
[3095] 1.00 0.78 1.00 0.78 1.00 1.00 0.25 0.67 1.00 1.00 1.00   NA   NA 1.00
[3109] 1.00 1.00 1.00 1.00 1.00 1.00 0.75 0.96 0.90 1.00 1.00 1.00 1.00 1.00
[3123] 1.00 1.00 0.96 1.00 0.00 1.00 1.00 1.00 1.00   NA 0.90 0.90 0.96 0.50
[3137] 1.00 1.00 1.00 0.80 1.00 0.33 0.23 0.90 0.17 1.00 1.00 1.00 1.00   NA
[3151] 0.90 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00   NA   NA 0.96 0.10
[3165] 1.00 1.00   NA 1.00 1.00   NA 0.60 1.00   NA 1.00 1.00 1.00 0.60 1.00
[3179] 0.60 0.50 0.60 0.96 1.00 1.00 0.96 0.96 1.00 1.00 1.00 1.00 1.00 1.00
[3193] 1.00 1.00 1.00 1.00   NA 1.00 0.77 1.00 1.00 0.96 0.96 1.00 1.00 1.00
[3207] 1.00 0.90 0.96 0.63   NA   NA 1.00 0.23 1.00 1.00 0.90 0.33 1.00 1.00
[3221] 1.00 1.00 1.00 1.00 0.40 0.95 1.00   NA 1.00 0.80   NA 0.95 0.70 0.00
[3235] 0.91 1.00 1.00   NA   NA   NA 0.98 1.00 1.00 1.00 1.00   NA 1.00   NA
[3249] 1.00 1.00 0.20 0.33 1.00 1.00 0.00 1.00 1.00 0.00 1.00 0.80 0.00 1.00
[3263] 1.00 1.00 0.00   NA 1.00 0.42 1.00 1.00 0.00 0.88 1.00   NA 0.50 1.00
[3277] 1.00 1.00 0.96 1.00 1.00   NA 1.00 1.00   NA   NA 1.00 1.00 1.00 1.00
[3291] 1.00 0.80 1.00 0.50 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.00   NA   NA
[3305] 1.00 1.00 0.96 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.40 0.40 0.40 0.40
[3319] 0.40   NA 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.80 1.00 1.00 0.83
[3333] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.60 1.00 1.00 0.10 1.00 1.00
[3347] 0.96 1.00 1.00 1.00 0.96 1.00 1.00 0.86 1.00 1.00 1.00 1.00 1.00 1.00
[3361] 0.96 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.96 1.00 1.00 0.17 1.00   NA
[3375] 0.95 1.00 1.00 1.00   NA 0.70 1.00 1.00 1.00 1.00 1.00 1.00 0.57 1.00
[3389] 1.00   NA 1.00 1.00 0.75 1.00 0.00 0.83 0.20 1.00 0.80 1.00 1.00 0.43
[3403] 1.00   NA 1.00 0.96 1.00   NA 1.00 1.00 1.00 0.75 1.00   NA 1.00 1.00
[3417] 1.00   NA 1.00 1.00 0.33 1.00 1.00 0.90 1.00 1.00 0.96 0.96 0.96 0.00
[3431] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.50 1.00 0.80 1.00 1.00
[3445] 1.00 1.00 0.88 1.00   NA   NA 1.00 1.00 1.00 1.00 1.00 1.00 0.80   NA
[3459] 1.00 0.89 1.00 1.00 1.00 1.00 0.83 1.00 0.75 0.00 1.00 1.00 0.82 1.00
[3473] 0.90 1.00 1.00 1.00 1.00 1.00 0.75 0.90 1.00   NA 1.00 0.96 0.83 0.50
[3487] 1.00 1.00 0.50   NA 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.90 1.00
[3501] 1.00 1.00 1.00 1.00 1.00 0.50 1.00   NA 0.90 1.00 1.00 0.93 0.70 1.00
[3515] 0.90 1.00 1.00 1.00 0.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00
```

```
[3529] 1.00   NA 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[3543] 0.88 1.00 1.00 1.00 1.00 1.00 0.80 1.00   NA 1.00 1.00 0.33 1.00 1.00
[3557] 1.00 1.00 1.00 0.70 1.00   NA   NA 1.00   NA 0.40 1.00 1.00 1.00 1.00
[3571] 0.90 0.70 1.00 0.83 1.00 1.00 0.70 1.00 1.00 1.00 0.70 1.00 1.00 1.00
[3585] 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 0.50 1.00 0.67 1.00 1.00 1.00
[3599] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00 0.86   NA 1.00 1.00
[3613] 1.00 1.00 0.90 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 1.00 0.80
[3627] 1.00 1.00 1.00   NA 1.00 1.00 1.00 1.00 0.80 0.40 1.00 1.00   NA 0.99
[3641] 1.00 1.00 0.80 1.00 1.00 1.00 0.80 1.00 1.00 1.00 1.00 1.00 0.90 1.00
[3655] 0.75 0.90 1.00 1.00 0.23 1.00 1.00 1.00 1.00 1.00   NA 1.00 1.00 1.00
[3669] 1.00 1.00 0.95 1.00 0.36 1.00 1.00 0.80 1.00 1.00 1.00 0.83 1.00 0.60
[3683] 0.00 0.90 1.00 1.00 1.00 0.88 0.00 1.00 1.00   NA 0.50 1.00 1.00 1.00
[3697] 0.67 1.00 1.00 0.90 0.60 0.90 1.00 1.00 1.00 1.00 0.90 1.00 0.90 0.90
[3711] 0.90 0.90 0.90 0.90 1.00 1.00 0.90 0.90 1.00 1.00 1.00 1.00 0.90 1.00
[3725] 0.80 0.96 0.70 1.00 1.00 1.00 0.77 1.00 1.00 0.80 1.00 1.00 1.00   NA
[3739] 0.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.70 0.40 1.00   NA   NA
[3753] 0.95 1.00 1.00 1.00 1.00 1.00 0.38 1.00 0.38 1.00 1.00 1.00 1.00 0.90
[3767]   NA 0.60 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.75
[3781] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.95   NA 1.00 1.00
[3795] 1.00 1.00   NA 1.00 1.00 0.91 1.00 1.00   NA 1.00 1.00 0.93 0.97 1.00
[3809] 0.75 0.91   NA 0.98 1.00 0.87   NA 1.00 0.70 0.90 1.00 1.00 1.00 0.80
[3823] 1.00 0.40 1.00 1.00 1.00 1.00 1.00 0.71 1.00 0.89 1.00 1.00 0.90 1.00
[3837] 1.00 0.80 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00   NA 1.00   NA
[3851] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.96 0.97 0.97 1.00   NA 1.00
[3865]   NA 1.00 0.70   NA 0.96 1.00 1.00   NA 1.00   NA 1.00 1.00
```

```r
d.airbnb_listings_feat$host_acceptance_rate <- as.numeric(sub("%", "", d.airbnb_listings_feat$host_accep
```

```
Warning: NAs durch Umwandlung erzeugt
```

```r
print(d.airbnb_listings_feat$host_acceptance_rate)
```

```
  [1] 1.00   NA 0.79 1.00 1.00 1.00 0.94 0.91 0.96 0.62 0.50 0.81 0.50 0.50
 [15] 0.81 1.00 0.94 1.00 0.64 0.92 0.93 0.00 1.00 1.00 0.00 1.00 0.71 0.00
 [29] 0.61 0.96 1.00 0.00 0.09 1.00 0.91 1.00 0.45 1.00 0.88 0.67 1.00 0.67
 [43] 0.94 0.94 0.94 0.94 0.52 0.94 0.94 0.94 0.94 0.94 0.94 0.94 0.52 0.94
 [57] 0.96 1.00 0.00 0.90 1.00 0.94 0.73 0.40 1.00 1.00   NA 0.33 0.75 0.80
 [71] 0.83 0.94   NA 0.93 1.00 0.95 0.00 0.40   NA 0.67 0.77 0.70 0.80 0.50
 [85] 0.80 0.50 0.67 0.85 1.00 1.00 1.00 1.00 0.63 0.33 0.62 1.00 0.84 0.63
 [99] 0.97 1.00 0.97 0.97 1.00 0.96 0.50 0.98 1.00 0.78 0.00 0.64 0.77 0.62
[113] 0.67 0.00 0.62 0.67 0.62 0.93 0.00 0.75 0.96 0.50 1.00 1.00 1.00 0.89
[127] 0.00 0.84 0.78 0.94 1.00 0.67 0.67 0.85 0.40 0.89 0.64 1.00 0.88 0.00
[141] 0.78 0.75 1.00 0.00 0.92 1.00   NA 0.99 1.00   NA 0.75 0.47 1.00 0.95
[155] 0.61 0.50   NA 0.56 0.79 0.78 0.53 0.90 0.00 0.29 0.80 0.70 1.00 0.53
[169] 0.53 0.53 0.53 0.62 0.67 1.00 0.97 1.00 0.00 0.66 0.25 0.24 0.87 1.00
[183] 0.62 0.60 0.25 0.97 0.81 0.08   NA 0.83   NA 0.29 0.47   NA 0.43 0.94
[197] 0.82 1.00 1.00   NA 0.62 0.00 1.00 0.75 0.75 0.75   NA 0.90 1.00 0.40
[211] 0.78 0.59 0.50 0.94   NA 1.00   NA 0.67 1.00 1.00 1.00 0.89 0.20 0.98
[225] 1.00 1.00 1.00   NA 0.92 0.38 0.50 1.00 0.90 0.73 1.00 0.80 0.44 1.00
[239]   NA 0.99 0.60 0.92   NA 0.00 0.80 1.00 0.38 0.68 0.60 0.00 0.96 0.00
[253]   NA 0.96 0.97 0.86 0.99 0.95 0.91 0.75 0.92 0.60 0.75 0.09 0.85 0.89
[267] 1.00 0.00 0.80 0.80 0.30 0.75 1.00   NA 0.00 1.00   NA 1.00 0.92 0.00
[281] 0.93 0.76 0.00 0.00 1.00 1.00 0.99 1.00 0.30 0.79 1.00 0.96 0.86 0.68
[295] 0.41 0.77 0.50 0.89 0.89 0.00 0.50 0.00 0.71 0.33 0.84 0.70 0.70 0.70
[309] 0.60 0.77 0.76   NA 0.50   NA 0.31 0.33 1.00 0.99 0.84 0.50 0.62 0.87
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 [323] 0.93 0.00 1.00 0.96 1.00 0.83 0.25 1.00 0.00 0.73 1.00 0.77 1.00 0.87
 [337] 0.87 0.83 1.00 1.00 0.50   NA 0.00 0.00 0.93 1.00 0.83   NA   NA 0.96
 [351] 0.96 0.96 0.96 0.96 0.96 0.99 1.00 0.47 0.79 1.00   NA 0.60 0.96 0.96
 [365] 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 1.00 0.96 1.00
 [379] 0.97 0.80 0.97 0.99 1.00 0.97 1.00 0.98 0.62 1.00 0.87   NA 0.62 0.60
 [393] 0.33   NA 0.93 0.00 0.86 0.70 0.92 0.00 1.00 0.95 1.00 1.00 0.88 0.88
 [407] 0.99 0.99 0.97 0.87 1.00 0.83 0.83 1.00 0.25 0.83 1.00 0.55 1.00 0.96
 [421] 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96
 [435] 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 1.00 0.96 0.96
 [449] 0.96 0.96 0.96 0.96 0.96 0.96 0.60 0.53 0.53 0.84 0.92 1.00 0.53 0.53
 [463] 0.53 1.00 1.00 0.53 0.80 0.94 0.94 0.94 1.00 0.86 0.00 0.92 0.92 0.92
 [477] 0.90 0.97 0.90 0.90 0.90 0.79 0.69 0.73 0.80 0.94 1.00 1.00 0.71 1.00
 [491] 1.00 1.00 1.00 0.81 0.98 1.00 0.32 1.00 0.80 0.97 0.50 0.88 1.00 1.00
 [505] 1.00 0.57 0.97 0.67 1.00 0.90 0.00 0.60 1.00 1.00 0.90 0.50 0.33 0.90
 [519] 1.00 1.00 0.00 1.00 1.00 0.60 0.92 0.94 1.00 1.00 0.96 0.70 0.70 0.70
 [533] 0.70 0.84 0.86 0.50 0.50 1.00 0.72 0.50   NA 1.00 1.00 1.00 0.55 1.00
 [547] 1.00 0.97 1.00 1.00 0.79 0.00 1.00 0.90 1.00   NA 1.00 0.60 0.00 1.00
 [561] 0.80 1.00 0.00 1.00 0.40 1.00 0.73 0.24 1.00 1.00   NA 0.23 1.00 0.97
 [575] 1.00 1.00 0.95 1.00 0.90 0.92 0.78 0.96 0.92 1.00 0.91 1.00 1.00   NA
 [589]   NA 0.62 1.00 0.92 0.84 0.00 0.71 0.62 0.60 0.80   NA 1.00 0.60 0.92
 [603] 0.88 0.91 1.00 0.94   NA 0.90 0.75   NA 1.00 0.50 0.97 0.96 1.00   NA
 [617] 0.97 0.50 0.84 0.98   NA 0.93   NA   NA 1.00 1.00   NA 1.00 1.00 0.47
 [631] 0.90 1.00 0.96 0.96 0.96 0.91 0.95 1.00 0.31 0.11 1.00 0.50 1.00 0.84
 [645] 0.98 1.00 0.00 1.00 1.00 0.60   NA 0.50 0.52 1.00   NA   NA 0.52 1.00
 [659] 0.83 0.87 0.86 0.00 0.97 0.00 0.67 0.90 0.47 0.00 0.67 1.00 0.90 0.68
 [673] 0.33 0.96 0.99 1.00 0.71 1.00 0.88 0.94   NA   NA 1.00   NA 0.80 1.00
 [687] 0.30 0.80 0.29 1.00 0.78 0.98 0.98 0.98 1.00 0.33 0.47   NA 1.00 1.00
 [701] 0.96 0.96 0.90 0.32 0.92 0.00 0.78 1.00 1.00 1.00 0.90   NA 0.88 0.95
 [715] 0.00   NA 0.89 1.00 0.50 0.92 1.00 1.00 0.40 1.00 0.98 0.98 0.98 0.98
 [729] 1.00 1.00 1.00 0.75 1.00   NA 0.96 0.70 0.76 0.24 1.00   NA   NA 0.83
 [743] 1.00 1.00 0.00 0.75 0.90   NA 0.71 0.71 0.33 1.00 0.96 1.00 0.00 1.00
 [757] 0.50   NA 0.77 0.00 0.87 1.00 1.00 0.00 1.00 1.00   NA 1.00   NA 0.91
 [771] 1.00 1.00 1.00 1.00 0.99   NA 0.97 1.00 0.80 0.24 0.87 0.33 1.00   NA
 [785]   NA 1.00   NA 0.81 0.94 0.50   NA 1.00 1.00 0.92 0.20 0.96 0.96 0.96
 [799] 0.96 1.00 0.00 0.00 0.83 1.00 1.00 0.00 0.70 0.99 1.00 1.00 1.00   NA
 [813] 0.64 1.00 0.50 0.91 0.94 1.00 1.00 0.88 0.87 1.00 0.99 0.88 1.00 1.00
 [827] 1.00   NA 0.33 1.00 0.98 1.00   NA 0.00 0.93 0.78 0.62 1.00 0.78 0.00
 [841] 0.60 0.10 1.00 1.00 0.58 0.56 0.43 0.99 0.55 1.00 1.00 0.00 0.41 0.80
 [855] 0.00 1.00 0.78 0.00 0.88 0.96 1.00   NA 0.44 0.50 1.00 0.97 0.62   NA
 [869]   NA 0.33   NA 0.00 1.00   NA 1.00 1.00 1.00 0.82 0.00 0.83 1.00 0.72
 [883] 0.86 0.92 1.00 0.93 0.93 0.83 1.00 1.00 0.70 0.00 1.00 0.34 0.40   NA
 [897] 0.96 0.00 0.25 0.93 1.00 1.00 1.00 1.00 0.00 1.00 0.50   NA 1.00 1.00
 [911] 0.55 0.33   NA 0.75 1.00 0.20 0.33 1.00 0.86 1.00 0.77 0.45 0.75 1.00
 [925] 0.91 1.00 1.00 0.50 0.67 0.94 1.00 1.00 1.00 0.94 0.33 0.93 1.00 0.23
 [939] 0.82 1.00 1.00 1.00 1.00 0.85 0.95 0.93 0.74 1.00 0.86 1.00 0.74 0.96
 [953] 1.00 0.83 1.00 1.00 0.70 0.67 0.00 0.00 1.00 0.50 0.33 1.00 0.60 0.97
 [967] 1.00 0.94 1.00   NA   NA 0.91 0.95 0.78 1.00 0.67 0.86 0.74 0.92 1.00
 [981] 0.93   NA 1.00 0.93   NA 0.74 0.50 1.00 1.00 0.98 1.00 0.67 0.95 0.00
 [995] 0.00 1.00 0.74 0.74 0.79 0.98 0.98 0.98 0.98 0.98 0.98 0.98 0.98 0.98
[1009] 0.98 0.92 1.00 0.60 1.00 1.00 1.00 0.40 0.93 0.20 0.47 0.93 0.00 0.93
[1023] 0.00 0.93 1.00 1.00 1.00 0.86 0.00 1.00   NA 1.00 0.94 1.00 1.00 1.00
[1037] 1.00 1.00 1.00 0.93 1.00 0.98 0.88 1.00 1.00 0.98 0.94 0.94 0.94 0.86
[1051] 1.00 1.00   NA 1.00 1.00   NA 0.93 1.00 1.00 1.00 1.00 1.00 1.00 0.00
[1065] 0.88 0.67 1.00 1.00 1.00 1.00 0.80 1.00 0.67 0.93 0.56 0.93 1.00 1.00
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[1079] 1.00 0.00 0.00   NA 0.98 0.92   NA   NA 0.40 0.40 0.97 0.88 1.00 1.00
[1093] 0.00 1.00 0.86 1.00 0.25 1.00 1.00 1.00 0.94 0.90 0.98 0.90 1.00 0.86
[1107]   NA 1.00 0.00 0.00 0.00 1.00 1.00 0.96 0.50 0.60 0.98 0.60 0.90 1.00
[1121] 1.00 0.96 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.92 0.92 1.00 1.00 0.60
[1135]   NA 0.17 0.74 1.00   NA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1149] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.96   NA 1.00 1.00 1.00 0.37
[1163] 1.00 0.60 0.83 0.62 0.60 0.62 1.00 1.00 0.98 0.60 1.00 1.00 1.00 1.00
[1177] 1.00 1.00 1.00   NA   NA 1.00   NA 0.70 0.93 1.00 0.50 1.00 0.67 1.00
[1191] 0.95 0.33 1.00 0.98 0.98 1.00 0.78 1.00   NA 0.95 0.33 1.00 0.00   NA
[1205] 0.75 1.00 1.00 1.00 0.58 0.50 0.70 0.00 1.00 1.00 0.92 1.00 0.62 0.50
[1219] 1.00 1.00 0.97   NA 0.67 0.83 0.53 1.00 1.00 1.00 1.00 1.00 1.00 1.00
[1233] 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.67 0.93 0.93
[1247] 0.92 0.00   NA 1.00   NA 1.00 0.00 1.00 1.00 1.00 0.98 0.93   NA 0.96
[1261] 0.22 1.00 0.93 0.00 0.83 0.43 1.00 1.00 0.33 0.00 1.00 0.96 0.96   NA
[1275] 1.00   NA 1.00 0.88 0.98 0.70 0.94 0.95 1.00 1.00 0.70 0.67 0.98 1.00
[1289] 0.81 1.00   NA   NA 1.00 1.00 0.94 0.94 0.98 0.91   NA 0.95 0.80 1.00
[1303] 0.86   NA 0.33 0.94   NA 1.00   NA 1.00 0.90 1.00 0.92 0.96 0.90 0.92
[1317] 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 1.00 1.00 0.91
[1331] 0.90   NA 1.00   NA 0.91 1.00 0.94 0.95 0.95 0.60 1.00 1.00 1.00 0.00
[1345] 0.95 1.00 0.71 1.00   NA 0.91 1.00 0.98 0.78 1.00 1.00 0.97 1.00 1.00
[1359] 0.00 1.00 0.93 0.97 0.97 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.80
[1373] 1.00 1.00 0.98 1.00 0.90 0.98 0.00 0.85 0.00 0.86 1.00 0.95   NA 0.50
[1387] 1.00 0.00 1.00 0.88 0.00 0.98   NA 0.71   NA 1.00 1.00 0.00 0.07 0.88
[1401] 0.00 0.33   NA 0.00 0.83 0.60 1.00 1.00 1.00 0.70 0.95 0.99 1.00 0.74
[1415] 0.00 0.98 0.40 0.91 1.00 0.93 0.96 0.78 1.00 1.00 1.00 0.91 1.00 1.00
[1429] 0.93 0.60 0.07 1.00 0.66 0.86 0.61 1.00 0.61 0.00 0.62 0.98 0.98 0.75
[1443] 0.96 0.70   NA 0.61 0.60 0.08 1.00 0.89 0.94 0.87 1.00 0.89 0.78 0.93
[1457] 1.00 1.00 1.00 1.00 1.00 0.96 0.96 0.96 0.96 0.93 1.00 1.00 1.00 1.00
[1471] 0.50 1.00 1.00 0.55 0.92 1.00 1.00 1.00 0.66 0.75 0.71 0.82 0.97 0.97
[1485] 0.62   NA 0.92 0.92 1.00 0.97 0.93 1.00 1.00 1.00 0.67 1.00 1.00 0.75
[1499] 0.96 1.00 0.14 1.00 1.00 0.96 0.50 0.79 0.75 0.92 0.00 1.00 1.00 0.80
[1513]   NA 0.97 0.83 0.60 0.88 1.00 0.26 1.00 1.00 0.99 0.26 1.00 0.77 1.00
[1527] 1.00 0.97 0.76 0.00 1.00 0.97 0.00 0.95 0.00 0.75 1.00 1.00 1.00 0.98
[1541] 0.50 1.00 1.00 1.00 0.25 1.00 0.94 0.94 0.94 0.38 1.00 1.00 0.99 0.92
[1555] 1.00 0.88 1.00 0.82 1.00 0.50 1.00 0.95 0.68 0.89 0.97 0.60 1.00 0.93
[1569] 0.93 1.00 0.85 1.00   NA 1.00 1.00 0.78 0.89 0.94 1.00 0.92 1.00 0.92
[1583] 0.97 1.00 0.96 0.92 0.92 0.92 0.78 0.00 0.96 1.00 0.99 0.96 0.00   NA
[1597] 0.75 1.00 1.00 0.93   NA 1.00 1.00   NA 1.00 1.00 1.00 1.00   NA   NA
[1611] 0.88 0.98 0.92 0.67 0.83 1.00 0.68 1.00 1.00 0.98 0.83 0.94   NA 1.00
[1625] 0.88 0.00 0.41 0.92 1.00 0.21 1.00 0.78 0.92   NA 0.92 0.29 1.00 0.99
[1639] 1.00 1.00 0.90   NA   NA 0.00   NA 0.25 1.00 0.50 0.50 1.00 0.92 1.00
[1653] 1.00 1.00 1.00 1.00 0.00 1.00 0.98 0.88 1.00 1.00 1.00 0.67 1.00 1.00
[1667] 0.67 1.00 1.00 0.96 0.40 0.96 0.96 1.00 0.67 1.00 0.80 0.43 0.75 1.00
[1681] 1.00 0.96 0.85 0.85 0.88 0.78 0.67 0.85 0.83 0.10 0.81 0.74 0.56 0.86
[1695] 0.97 1.00 1.00 1.00 0.92 0.92 0.92 0.92 0.92 0.92 1.00 1.00 1.00 0.25
[1709] 0.62 0.65 0.00 0.61 1.00 1.00 0.96 0.88 1.00 1.00 0.74 0.00 0.96 0.00
[1723] 1.00 0.75 0.96 0.00 0.78 1.00 1.00 1.00 1.00 1.00 0.32 0.59 0.00 0.60
[1737] 0.98 1.00   NA 0.88 0.00 1.00 0.98 0.96 0.33 0.89 0.50   NA 0.96 1.00
[1751] 0.43 1.00 0.96 0.98 1.00 1.00 0.95 0.40 0.22   NA 1.00 0.86   NA 1.00
[1765] 1.00 0.00 0.97 0.96 1.00 1.00 0.74 0.74 0.74 0.45 0.40   NA 0.80 0.96
[1779] 0.96 0.60 0.00 1.00 0.50 1.00 0.96 0.33 0.95 0.64 0.96 1.00   NA 0.00
[1793] 1.00 1.00 0.92 0.98 0.97 1.00 1.00 1.00 0.75 0.00 0.99 1.00 1.00 1.00
[1807] 0.43 0.99 0.00 1.00 1.00 0.96 0.94 0.77 0.83 0.95 0.97 0.96 1.00 0.98
[1821] 1.00 1.00 0.84 0.87 0.00 1.00 1.00 1.00   NA   NA 0.96 0.74 0.98 1.00
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[1835] 0.70 0.15 1.00 1.00 0.97 0.98 0.96 1.00 1.00 0.96   NA 0.75 1.00 1.00
[1849] 0.96 0.96 0.77 0.00 1.00 1.00 1.00 1.00 0.00 0.40 0.40 0.00 1.00 0.90
[1863] 0.96 0.60 0.96 0.81 0.83 0.67 0.60 0.25 0.60 0.67 1.00 0.65 0.73 0.67
[1877] 0.96 0.74 0.74 0.00 0.86 1.00 0.85 0.63 0.00 0.99 0.84 0.97 0.96 0.98
[1891] 0.97 0.97 1.00 0.98 0.98 0.97 0.91 1.00 0.97 0.97 0.91 0.79 0.25 0.50
[1905]   NA 1.00 0.50 0.75 0.50 0.33 0.62 0.50 0.78 0.78 1.00 0.88 0.23 0.77
[1919] 0.83 0.40 0.11 1.00 0.50 0.62 0.62 0.62 0.62 0.50 0.97 0.79 0.67 0.98
[1933] 1.00 1.00 1.00 1.00 1.00   NA 1.00 0.26 0.26 0.99 1.00 0.00 0.97   NA
[1947] 0.90 0.74 0.74 0.74 0.26 1.00 1.00 0.00 1.00 0.69 0.97 1.00 1.00 0.60
[1961] 1.00 0.97 0.62 1.00 0.98 0.50 1.00 1.00 1.00 1.00 0.00 0.96 0.90   NA
[1975] 0.96 0.94 0.74 0.61 0.47 0.40 0.95 0.60 0.98 1.00 0.71 1.00 0.96 0.00
[1989] 0.61 0.92 1.00 0.71   NA 0.83 0.67 1.00 0.64 1.00 0.00 0.67 0.75 0.00
[2003]   NA 0.97 1.00 0.67   NA 0.75 1.00 0.80 1.00 1.00 0.79 0.53 1.00 0.00
[2017] 0.94 0.29 1.00 1.00 0.93 0.50 1.00   NA 0.00 1.00 0.00 0.95 0.97 1.00
[2031] 0.96 1.00 1.00 0.92 0.75 1.00 1.00 1.00 0.50 1.00 1.00 1.00 1.00 0.33
[2045] 0.00 0.00 0.85 0.92 0.41 0.96 0.96 0.29 0.00 1.00 0.33 0.05 0.66 0.15
[2059] 0.83 1.00 0.85 1.00 0.96 0.00 0.75 0.46 0.83 0.81 0.78 0.82   NA 0.55
[2073] 0.60 0.85   NA 0.79 0.71 1.00 0.00 1.00 0.00 0.84 1.00 0.07 0.86 0.70
[2087] 0.09 0.97 0.00 0.98   NA 0.97 0.71 0.38 0.78 1.00   NA 0.56 0.98 0.00
[2101] 0.57 0.95 0.98 0.00 1.00 0.00 1.00 0.96 1.00 1.00 0.67 0.99 0.99 1.00
[2115] 1.00 1.00 1.00 0.85 1.00 0.86 0.00 0.00 0.74 0.98 0.75 1.00 0.97 0.00
[2129] 1.00 1.00 1.00 0.07 0.59 1.00 0.00 0.92 0.99 0.00 0.74 0.91 0.88 0.50
[2143] 0.00 0.83 0.91 0.61 0.99 0.95 1.00 0.78 0.98 0.00 1.00 0.96 0.15 0.92
[2157] 0.40 1.00 0.00 1.00 1.00 1.00 0.96 0.00 0.33 0.74 0.43 1.00 1.00 0.74
[2171] 0.60 0.88 0.74 1.00 1.00 0.50 0.67   NA 0.88 1.00 0.00 0.74 0.92 0.60
[2185] 1.00 1.00 0.96 1.00 1.00 0.92 1.00 0.74   NA 0.00 0.86 0.77 0.98 0.92
[2199] 0.67 1.00 0.96 0.75 0.96 0.27 0.23 0.14 0.41 1.00 1.00 0.57 0.85 0.97
[2213] 0.92 0.00 0.54 0.25 0.75 0.97 0.33 0.92 0.85 0.92 0.80 1.00 0.92 0.92
[2227] 0.00   NA 0.96 0.93 1.00 0.36 1.00 0.25 0.70 0.96 0.97 0.67 0.94 1.00
[2241] 1.00 1.00 0.77 0.99 1.00 0.90 0.30 0.30 1.00 0.14 0.71 0.47 1.00 0.61
[2255]   NA 1.00 0.98 0.74 0.13 0.92 0.67 0.99 1.00 0.75 1.00 1.00   NA 1.00
[2269] 0.96 0.70 0.98 0.00 0.00   NA 0.07 0.17 0.97 1.00 0.85 0.00 0.00 0.14
[2283] 1.00 0.50 1.00 0.66 1.00 0.96 0.98 0.52 0.35 0.98   NA 1.00 0.88 0.96
[2297] 0.26 1.00 1.00 0.33   NA 0.79 1.00 1.00 0.96 0.67 0.96 1.00 0.73 0.62
[2311] 0.96 0.85 1.00 1.00 1.00 1.00   NA 0.92 0.92 1.00   NA 1.00 0.89 1.00
[2325] 0.62 0.00 1.00 1.00 1.00 0.60 1.00 0.57 1.00 0.62   NA   NA 0.99 0.99
[2339] 0.75 1.00 0.98 0.98 0.74 0.74 0.80 0.38 0.50 1.00 1.00 1.00 0.99 0.20
[2353] 1.00 1.00   NA 0.78 0.44 0.94   NA 1.00 0.96 0.25 0.69 1.00 0.99 1.00
[2367] 1.00 1.00 1.00 0.78 0.96 0.99 0.99 0.71 1.00 0.97 1.00 1.00 0.87 0.87
[2381] 0.87 0.73 1.00 0.92 0.89 0.81 0.30 1.00 1.00 0.40 0.84 1.00 0.29   NA
[2395] 0.98 0.60 1.00 0.13 0.60 0.93 0.52 1.00 1.00 0.00 0.75 0.00 0.93 0.00
[2409] 0.96 0.52 0.74 0.00   NA 0.87 0.87   NA 0.92 0.91 0.98 0.96 0.43 1.00
[2423] 0.98 1.00 0.92 0.66 0.92 0.92 0.92 0.83 0.92 0.92 0.92 0.92 0.92 0.92
[2437] 0.92 0.92 0.92 0.92 0.92 0.98 0.05 1.00 1.00 1.00 0.78 0.80 1.00   NA
[2451] 1.00 0.86 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92
[2465] 0.81 0.92 0.96 0.38 0.69 1.00 0.92 0.74 0.92 0.92 0.92 0.92 1.00 0.99
[2479] 1.00 1.00 1.00 0.77 0.78 0.00 1.00 0.00 0.24   NA 0.62 0.60 1.00 0.22
[2493] 0.20 0.96 0.92 1.00 1.00 1.00 0.69 0.75   NA 0.85 0.85 0.85 0.50 0.50
[2507] 0.50 0.50 0.99   NA 0.38 1.00   NA 1.00 0.84   NA 0.85 0.90 0.83 0.00
[2521] 0.60 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.76 0.30 0.85
[2535] 0.00 0.99 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96 0.96
[2549] 0.96 0.96 0.96 0.96 1.00 0.88 0.43 0.00 0.94 0.94 0.02 1.00 0.96 0.97
[2563] 0.96 0.00 1.00 1.00 1.00 1.00 1.00 0.00 0.25 0.85 0.96 0.67 1.00 0.92
[2577] 1.00 0.50 1.00 0.74 0.74 0.50 1.00 0.11 0.75   NA 0.00 0.88 0.98   NA
```

```
[2591] 1.00 0.55 0.69 0.60 0.75 0.00 0.96 0.00   NA 0.00 1.00 1.00   NA 0.60
[2605] 0.88 0.00 1.00 1.00 0.17   NA 0.20 1.00 1.00 0.20 0.39 0.71   NA 0.91
[2619] 1.00 0.00 0.96 0.00 1.00 0.00 0.80 1.00 0.82 0.93 0.33 1.00   NA 0.67
[2633] 1.00 0.96 0.92 1.00 1.00 0.56 1.00 0.96 0.00 0.96 0.55 0.77 0.14   NA
[2647] 0.60 0.67 0.70 0.52 0.00 0.99 0.92 0.77 0.96 0.70 1.00 0.11   NA 0.97
[2661] 0.75 0.50 0.93 0.96 0.96 0.74 0.13 1.00 0.57 1.00 1.00 0.60 0.50 0.00
[2675] 0.50 1.00 0.00 1.00 0.93 0.85 0.86 0.95 0.50   NA 0.88   NA 0.71 0.95
[2689]   NA 0.88 0.65 0.92 0.00 1.00 0.90   NA 0.95 0.77 0.00 0.98 0.98 0.47
[2703] 1.00 0.76 1.00 0.00 0.50 1.00 0.86 0.00 0.85 0.00 1.00 0.74   NA 0.00
[2717] 0.86 0.00 0.74   NA 0.57 0.33 1.00 0.17 0.90 0.96   NA 0.50 0.73 0.00
[2731] 0.75 0.64 1.00 1.00 1.00 0.00 0.92 0.46 0.96 0.71 0.96 0.95 0.94 0.88
[2745] 0.37 0.77 1.00 0.05 1.00 1.00 0.00 0.52 0.88 0.23 0.00 0.89 0.97 1.00
[2759] 1.00 0.23 0.88 1.00 0.77 0.85 0.97 0.80 0.33 0.53 0.50 0.24 1.00 0.67
[2773] 0.63 0.74 1.00 0.31 0.97   NA 0.97 0.82 0.70 0.70 0.74 0.96 0.88 0.92
[2787] 1.00   NA 0.74 0.74 0.74 0.74 0.74 0.74 0.74 1.00 1.00 1.00 1.00 0.00
[2801] 0.41 0.00 0.00 0.00 0.44 0.53 1.00 1.00 0.07 1.00 0.22 0.92 0.84 0.00
[2815] 0.52 0.96 0.91 0.88 0.99   NA 0.70 0.63 1.00 0.75 1.00 0.00 0.10 0.94
[2829] 0.00 0.94 0.98 0.86 0.96 0.71 1.00 1.00 1.00 1.00 0.85 0.83 0.96 0.00
[2843] 0.00 0.88 0.78 0.96   NA   NA 0.59 0.00 0.74 1.00 0.98 0.19 0.00 0.92
[2857] 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92
[2871] 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.92 0.00   NA 0.75 0.95
[2885] 0.67 0.74 0.87 0.85 1.00 0.74 0.99 0.74 0.74 0.85 1.00 0.14 0.70 0.91
[2899] 1.00 0.00 0.98 0.00   NA 0.94 0.00 0.77 0.90 1.00 0.00   NA 1.00 0.30
[2913] 1.00 1.00 1.00 0.82 0.82 0.08 0.75 0.92 1.00 0.98 0.67 0.99 0.95   NA
[2927] 0.92 0.80 0.57   NA 1.00 1.00 0.92 0.98 1.00 1.00 1.00 0.60 0.73 0.82
[2941] 0.00 1.00   NA 0.83 1.00 1.00 0.00 0.22 0.77 0.70 0.62 0.64 0.96 0.78
[2955] 0.60 0.80 1.00 0.77 0.78 0.74 1.00 0.74 0.74 1.00 0.75 0.00 0.96 1.00
[2969] 0.50 0.97 0.64 1.00 0.54 0.97 1.00 1.00 1.00 1.00 1.00 0.98 1.00 1.00
[2983] 0.98 0.67 0.42 0.77 0.67 0.57 0.80 0.96 0.96 0.90 0.89 0.48 0.87 0.00
[2997] 1.00 1.00 0.60 0.73 0.95 0.89 0.74 0.54 0.64 0.97 1.00 1.00 0.86 0.56
[3011] 0.74 1.00 0.74 0.39 1.00 0.91 1.00 0.97 0.74 1.00 1.00 0.93 1.00 0.96
[3025] 0.97 1.00   NA 0.25 0.94 1.00 1.00 0.96 1.00 0.20 0.67 0.19 0.14 0.98
[3039] 0.33 0.33 0.33 1.00 1.00 0.74 0.83 1.00 1.00 0.83 1.00 0.00 1.00 1.00
[3053] 0.95 0.75 0.96 0.80 0.74 0.74 0.21 0.55 0.95 0.95 0.95 0.85 0.94 0.30
[3067] 1.00 0.98 0.85 1.00 1.00 1.00 0.00 0.70   NA 1.00 0.33 0.48 0.70 0.85
[3081] 0.85 0.98 1.00 0.43 0.30 1.00 1.00 0.31 0.72 1.00 0.52 1.00 0.74 0.52
[3095] 0.97 1.00 0.93 1.00 0.96 1.00 0.00 0.30 0.62 0.93 0.78 0.67 1.00 1.00
[3109] 0.71 1.00 0.96 1.00 0.93 0.93 0.74 0.74 0.50 1.00 1.00 1.00 1.00 1.00
[3123] 1.00 1.00 0.74 1.00 0.25 1.00 0.82 0.82 0.72 0.50 0.50 0.50 0.74 0.80
[3137] 0.96 1.00 1.00 0.70 1.00 0.00 0.09 1.00 0.00 1.00 0.58 0.62 0.95 0.00
[3151] 1.00 1.00 1.00 1.00 1.00 0.61 1.00 0.82 1.00 0.52 1.00   NA 0.85 0.09
[3165] 1.00 0.97 1.00 1.00 0.93 0.93 1.00 0.73 1.00 1.00 1.00 1.00 1.00 1.00
[3179] 1.00 0.33 1.00 0.85 1.00 0.87 0.85 0.85 0.96 1.00 1.00 0.88 0.56 1.00
[3193] 0.96 0.78 0.73 0.67   NA 1.00 0.99 0.96 0.62 0.85 0.85 0.96 0.96 1.00
[3207] 0.92 0.98 0.76 0.91 1.00 0.00 0.93 0.09 1.00 0.94 0.96 0.67 1.00 1.00
[3221] 1.00 1.00 1.00 1.00 0.83 0.97 0.83   NA 0.82 0.94 0.92 0.87 0.61 0.00
[3235] 0.97 0.52 1.00 0.40 1.00 1.00 1.00 0.86 0.71 0.77 0.91 1.00 0.81 1.00
[3249] 0.96 1.00 0.89 0.00 1.00 1.00 0.33 1.00 1.00 0.36 0.82 0.94 0.00 1.00
[3263] 0.48 1.00 1.00 0.00 0.76 0.72 1.00 0.94   NA 0.96 0.75   NA 0.20 0.60
[3277] 0.88 0.57 0.85 1.00 0.44   NA 1.00 0.92   NA   NA 0.99 0.44 0.90 0.65
[3291] 0.86 0.53 0.96 0.00 0.91 1.00 1.00   NA 1.00 0.83 0.81 0.07   NA   NA
[3305] 1.00 1.00 0.85 1.00 0.61   NA 0.50 1.00 1.00 0.67 0.83 0.83 0.83 0.83
[3319] 0.83   NA 0.96 0.96 0.64 0.87 1.00 0.50 1.00 0.84 0.00 1.00 0.84 0.55
[3333] 0.96 0.98 0.98 1.00 0.95 0.89 1.00 0.50 0.71 0.61 1.00 0.00 0.70 1.00
```

```
[3347] 0.85 1.00 1.00 1.00 0.85 1.00 1.00 0.72 0.92 1.00 1.00 0.50 1.00 0.96
[3361] 0.85 0.90 0.44 0.44 0.44 0.98 0.50 0.87 0.85 0.80 0.80 0.54 0.00 0.50
[3375] 0.97 1.00   NA 1.00 1.00 0.14 0.95 0.65 0.96 1.00 1.00 1.00 0.50 1.00
[3389] 0.97 0.50 1.00 0.84 0.20 1.00 0.06 0.86 0.00 0.95 0.74 0.29 1.00 0.45
[3403] 1.00   NA 1.00 0.74 0.95 0.50 1.00 0.84 1.00 0.20 0.98 0.00 0.99 1.00
[3417] 0.34 0.00 0.85 0.99 0.25 0.98 0.87 0.95 1.00 0.92 0.85 0.85 0.85 0.00
[3431] 0.77 0.67 1.00 0.54 0.98 0.92 0.95 0.00 0.80 0.17 0.71 0.25 0.38 1.00
[3445] 0.90 0.25 0.75 1.00 0.40   NA 0.70 0.67 1.00 0.96 0.96 0.80 0.14   NA
[3459] 1.00 0.77 0.50 0.75 0.25 1.00 0.86 0.96 0.33 0.00 1.00 0.00 0.69 0.52
[3473] 0.06 0.67 0.94 0.97 0.50 0.89 0.20 0.69 0.97   NA 0.50 0.85 0.46 0.18
[3487] 0.92 1.00 0.80   NA 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.68 1.00
[3501] 0.93 0.70 1.00 0.86 0.96 0.86 0.50   NA 1.00 0.99 1.00 0.86 0.59 0.38
[3515] 0.29 0.97 0.97 0.97 0.00 1.00 0.50 0.67 0.00 0.67 0.67 0.97 0.86 0.71
[3529] 0.91   NA 1.00 0.99 0.87 0.00 0.93 0.95 0.46 0.87 0.77 0.96 0.75 0.38
[3543] 0.33 0.67 0.50 1.00 0.83 0.92 0.21 1.00   NA 1.00 0.71   NA 1.00 0.10
[3557] 0.97 1.00 0.52 0.74 0.77 0.00   NA 0.93 0.00 0.83 0.50 0.60 0.67 0.79
[3571] 0.96 0.79 1.00 0.46 0.94 1.00 0.80 0.83 1.00 0.00 0.57 0.70 1.00 1.00
[3585] 1.00 1.00 1.00 0.86   NA 0.80 0.67 0.74 0.00 0.94 0.70 1.00 0.20   NA
[3599] 1.00 0.94 0.94 0.94 0.94 0.94 0.94 0.75   NA 1.00 0.58 1.00 0.99 1.00
[3613] 1.00 0.97 0.62 0.00 0.17 0.96 0.77 0.00 0.94 1.00 0.89 0.99 0.96 0.58
[3627] 0.40 1.00 0.00   NA 0.96 0.91 1.00 1.00 0.76 0.83 0.97 0.00   NA 0.87
[3641] 0.96 0.67 1.00 0.50 0.00 1.00 1.00 0.82 1.00 0.00   NA 1.00 1.00 0.69
[3655] 0.50 0.75 1.00 0.86 0.09 0.44 0.83 0.86 1.00 0.85   NA 1.00 1.00 1.00
[3669] 0.88 0.93 0.97 0.52 0.00 1.00 0.92 0.50   NA 0.67 1.00 0.46 0.92 0.00
[3683]   NA 0.75 0.95 0.97 0.67 0.96 0.00 0.00 1.00   NA 1.00 1.00 1.00   NA
[3697] 0.00 1.00 0.65 0.75 0.13 1.00 0.82 1.00 0.88 1.00 0.78 0.92 0.93 1.00
[3711] 1.00 1.00 1.00 1.00 0.83 1.00 1.00 0.46 0.96 1.00 0.80 1.00 0.46 0.56
[3725] 0.60 0.85 0.33 1.00 1.00 0.78 0.54 0.96 1.00 0.58 1.00 0.96 0.52   NA
[3739] 0.00 1.00 1.00 1.00 0.90 1.00 1.00 0.96 0.65 0.33 0.19   NA   NA   NA
[3753] 0.87   NA 0.91 1.00 1.00 1.00   NA 1.00   NA 1.00 1.00 0.71 0.98 0.52
[3767]   NA 1.00 1.00 0.98 0.00 0.75 1.00 1.00 1.00 0.75 0.81 0.00 0.00 0.33
[3781] 1.00 0.83 0.60 0.60 1.00 1.00 0.99 1.00 0.96 1.00 0.87   NA 1.00 0.97
[3795] 1.00 0.95   NA 1.00 1.00 0.94 0.98 1.00   NA 0.97 0.75 0.86 0.05 0.86
[3809] 0.57 0.94   NA 1.00 0.94 0.89   NA 1.00 0.79 0.98 1.00 0.97 1.00 0.77
[3823] 1.00 0.08 1.00 1.00 1.00 0.98   NA 0.65 1.00 0.39 1.00 1.00 0.94 0.99
[3837] 1.00 1.00 1.00   NA 1.00 0.99 1.00 1.00 0.50 0.99   NA   NA 1.00   NA
[3851] 1.00 1.00 1.00 0.70 1.00 1.00 1.00 1.00 0.47 1.00 1.00 0.80   NA 0.00
[3865] 1.00 0.00 0.00   NA 0.85 1.00   NA   NA 0.69   NA 0.86 1.00
```

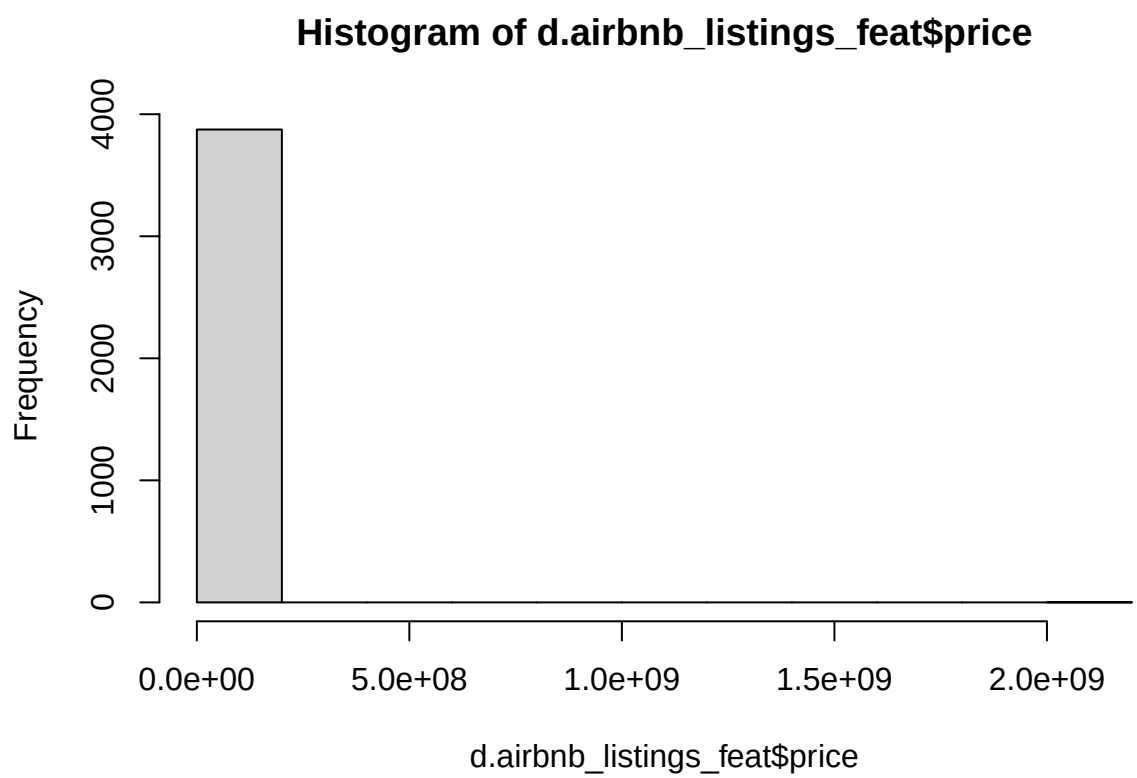
```r
str(d.airbnb_listings_feat)
```

```
'data.frame':   3876 obs. of  23 variables:
 $ host_response_time          : chr  "within an hour" "N/A" "within a few hours" "N/A" ...
 $ host_response_rate          : num  1 NA 1 NA 1 1 1 1 1 1 ...
 $ host_acceptance_rate        : num  1 NA 0.79 1 1 1 0.94 0.91 0.96 0.62 ...
 $ host_is_superhost           : num  0 0 1 0 1 1 1 1 1 1 ...
 $ host_verifications          : chr  "['email', 'phone']" "['email', 'phone']" "['email', 'phone']" "[
 $ host_has_profile_pic        : chr  "t" "t" "t" "t" ...
 $ host_identity_verified      : chr  "t" "t" "t" "t" ...
 $ neighbourhood_group_cleansed: chr  "Aigle" "Nyon" "Aigle" "Lausanne" ...
 $ latitude                    : num  46.4 46.5 46.3 46.5 46.5 ...
 $ longitude                   : num  7.13 6.33 7.06 6.64 6.8 ...
 $ property_type               : Factor w/ 63 levels "Barn","Boat",..: 18 44 18 16 52 18 18 44 11 18 ..
 $ room_type                   : Factor w/ 4 levels "Entire home/apt",..: 1 3 1 1 2 1 1 3 1 1 ...
 $ accommodates                : int  3 5 6 3 4 2 2 1 3 14 ...
```

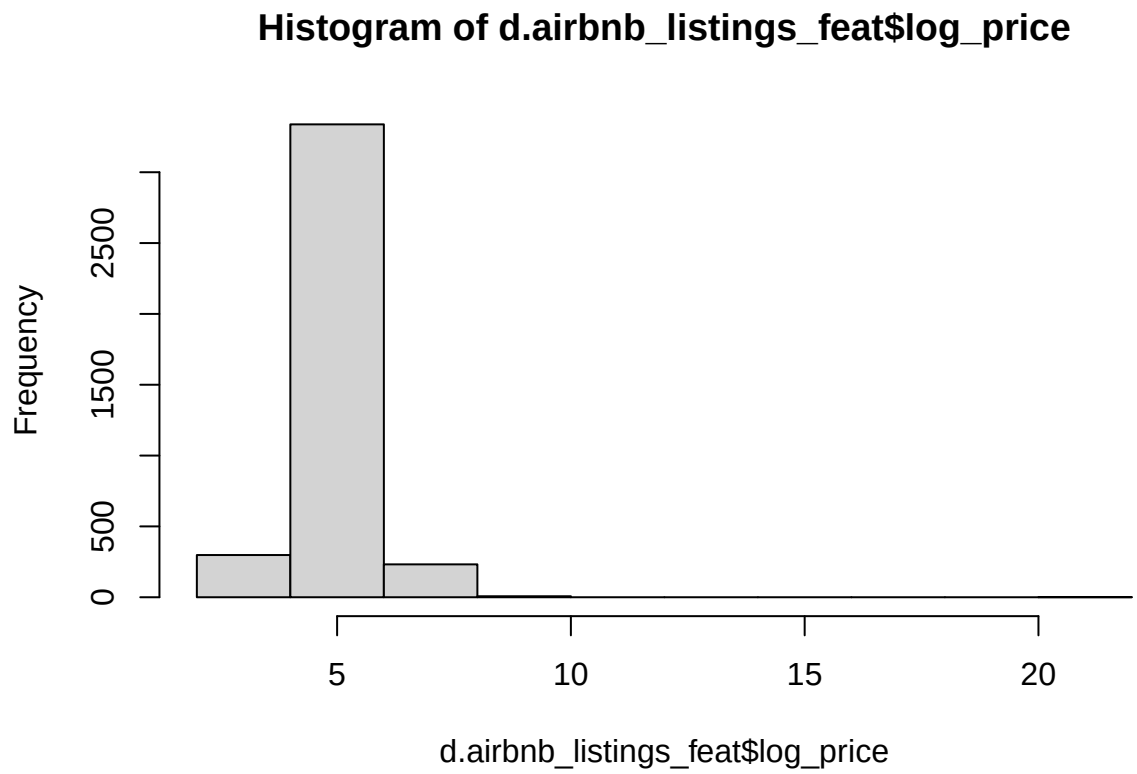
```
 $ bathrooms                   : num  1 NA 3 1.5 1 1 1 1 1 2.5 ...
 $ beds                        : int  2 4 4 1 3 2 1 1 2 10 ...
 $ price                       : num  107 725 590 144 254 277 101 34 127 278 ...
 $ minimum_nights              : int  3 4 4 3 1 1 5 2 2 2 ...
 $ number_of_reviews           : int  70 0 18 22 85 125 26 292 41 197 ...
 $ estimated_occupancy_l365d   : int  42 0 0 0 42 120 0 78 150 78 ...
 $ review_scores_rating        : num  4.74 NA 4.94 4.68 4.75 4.87 4.23 4.67 4.95 4.68 ...
 $ log_price                   : num  4.67 6.59 6.38 4.97 5.54 ...
 $ price_per_guest             : num  35.7 145 98.3 48 63.5 ...
 $ high_demand                 : num  0 0 0 0 0 0 0 0 0 0 ...
```

Exploratory Data Analysis

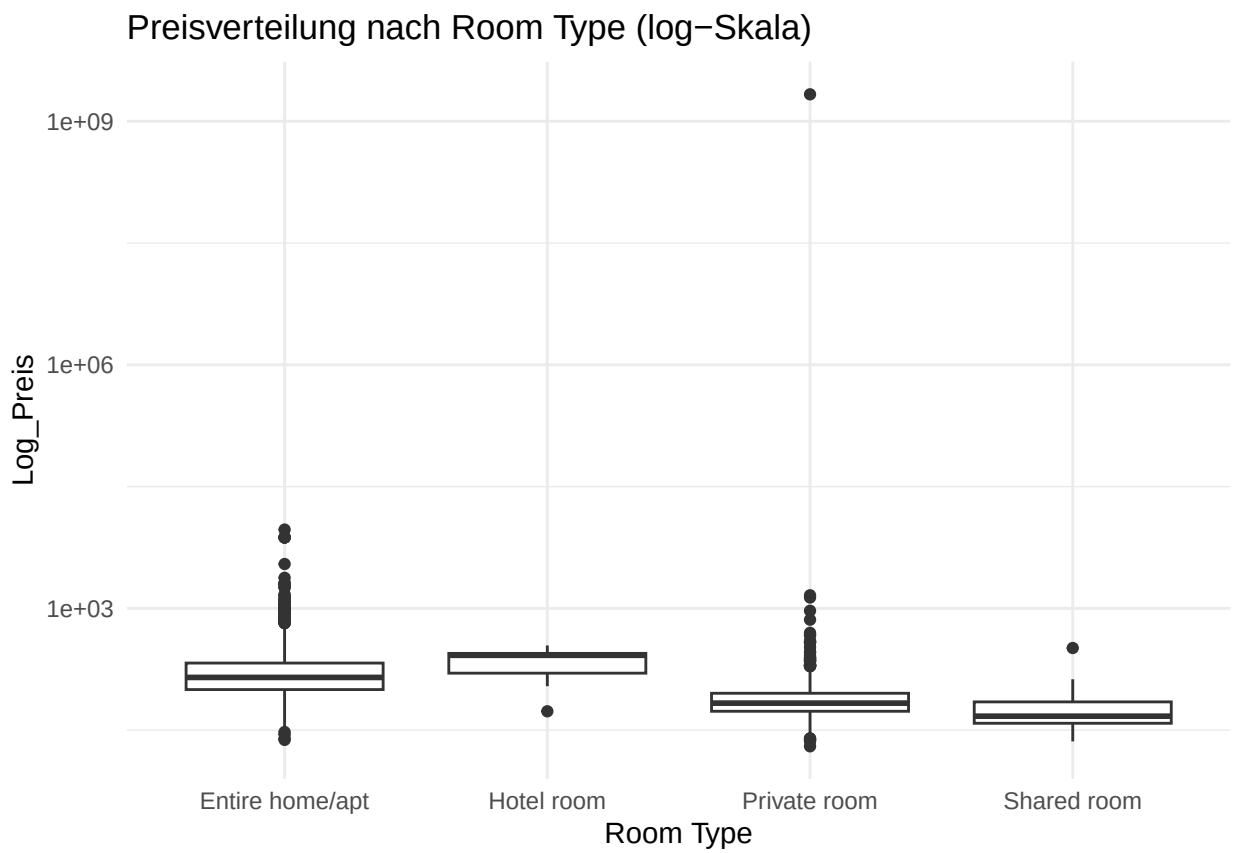
```r
hist(d.airbnb_listings_feat$price)
```




```r
hist(d.airbnb_listings_feat$log_price)
```

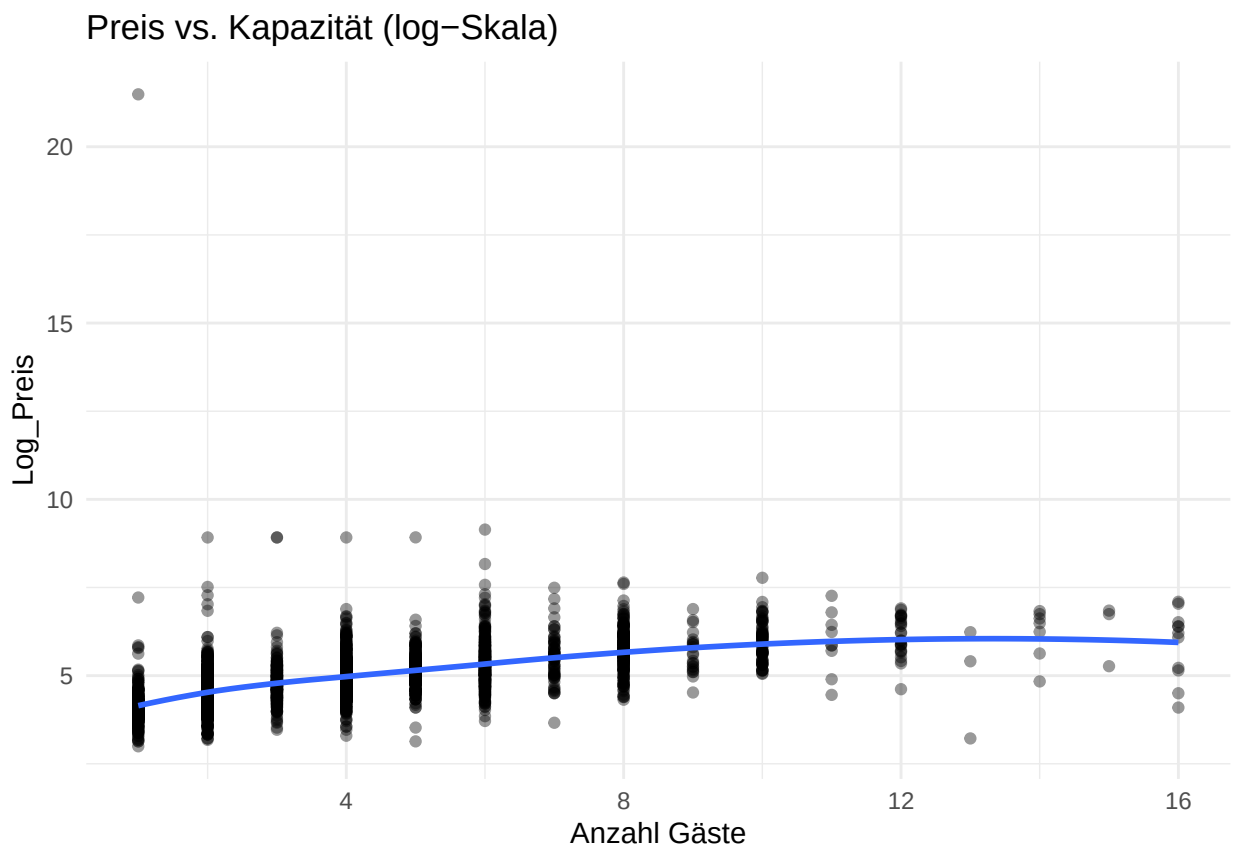



```r
ggplot(data= d.airbnb_listings_feat, aes(x = room_type, y = price)) +
  geom_boxplot() +
  scale_y_log10() +
  labs(
    title = "Preisverteilung nach Room Type (log-Skala)",
    x = "Room Type",
    y = "Log_Preis"
  ) +
  theme_minimal()
```



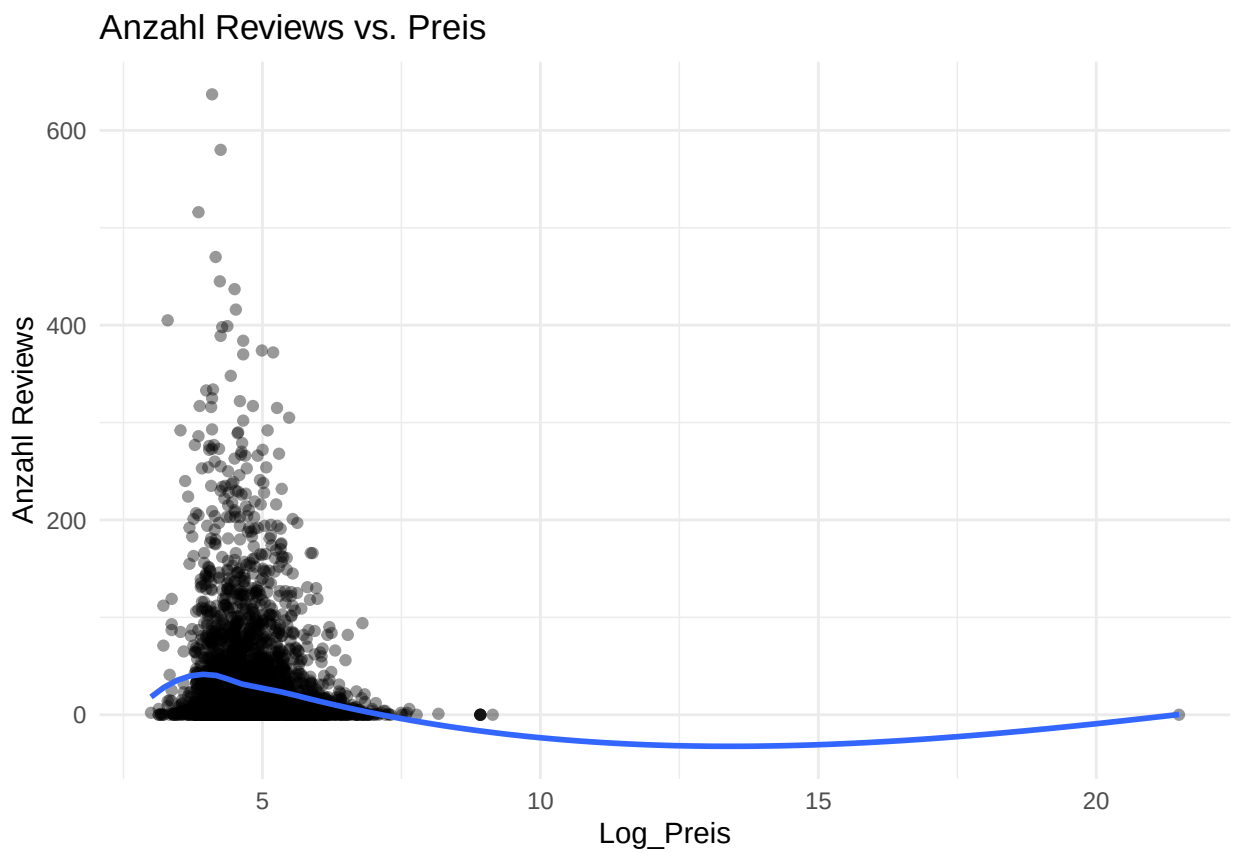

```r
ggplot(data = d.airbnb_listings_feat, aes(x = accommodates, y = log_price)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Preis vs. Kapazität (log-Skala)",
    x = "Anzahl Gäste",
    y = "Log_Preis"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

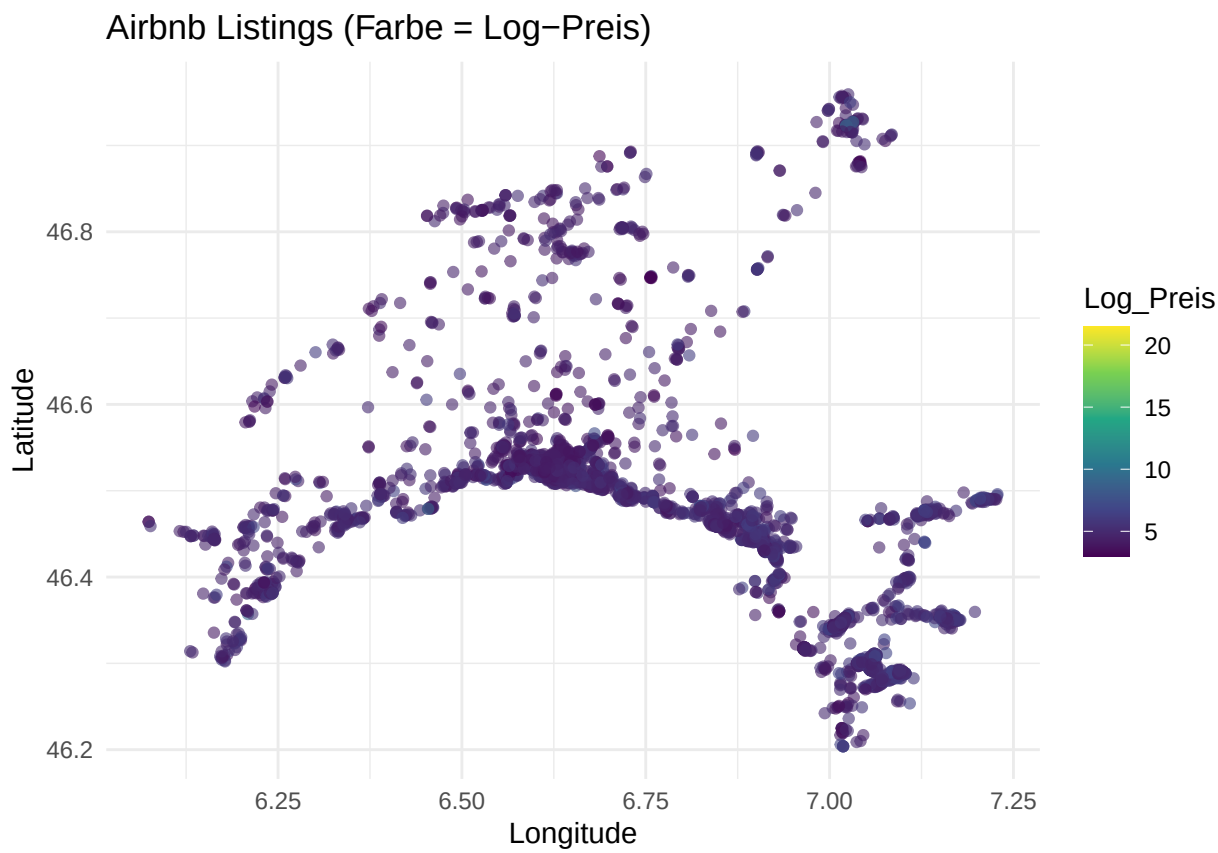



```
ggplot(data = d.airbnb_listings_feat, aes(x = log_price, y = number_of_reviews)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Anzahl Reviews vs. Preis",
    x = "Log_Preis",
    y = "Anzahl Reviews"
  ) +
  theme_minimal()
```

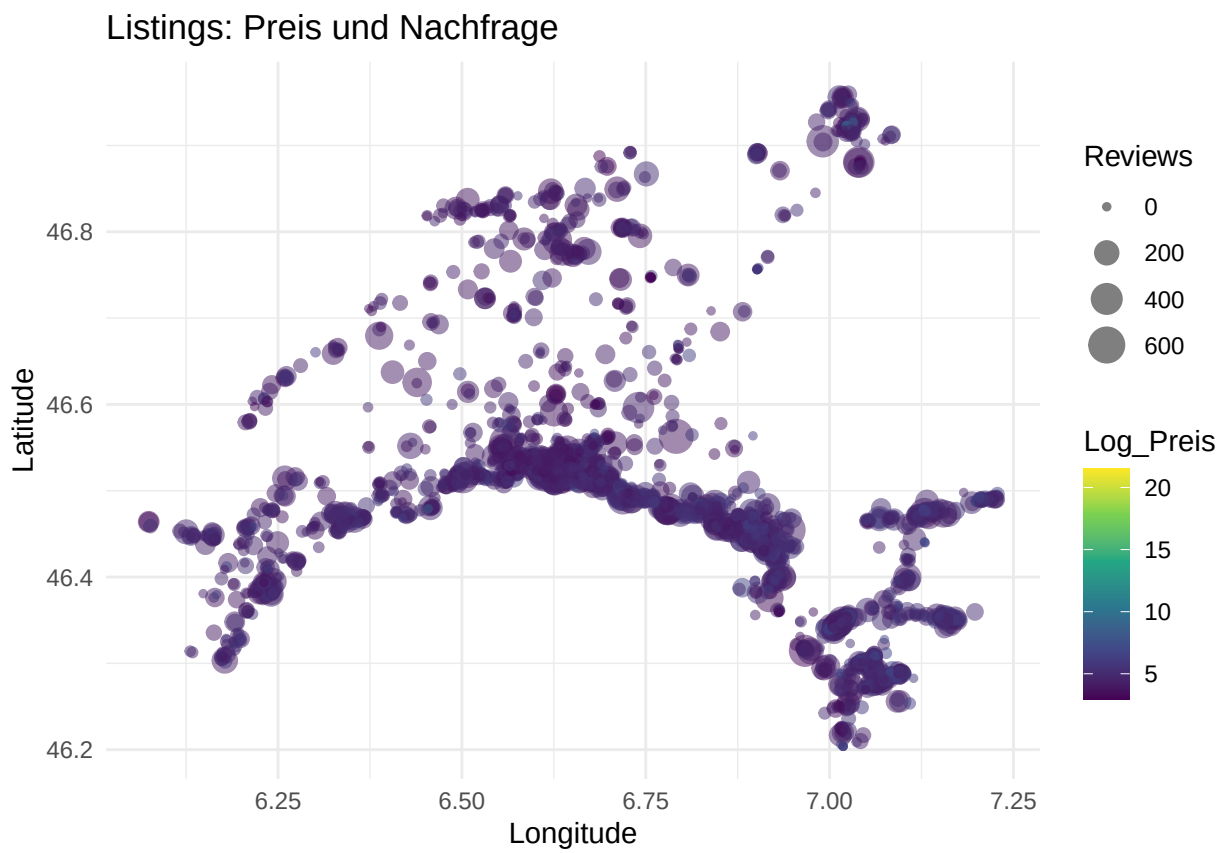
```
`geom_smooth()` using formula = 'y ~ x'
```




```r
ggplot(data = d.airbnb_listings_feat, aes(x = longitude, y = latitude)) +
  geom_point(aes(color = log_price), alpha = 0.6) +
  scale_color_viridis_c() +
  labs(
    title = "Airbnb Listings (Farbe = Log-Preis)",
    x = "Longitude",
    y = "Latitude",
    color = "Log_Preis"
  ) +
  theme_minimal()
```




```r
ggplot(data = d.airbnb_listings_feat, aes(x = longitude, y = latitude)) +
  geom_point(aes(color = log_price, size = number_of_reviews), alpha = 0.5) +
  scale_color_viridis_c() +
  labs(
    title = "Listings: Preis und Nachfrage",
    x = "Longitude",
    y = "Latitude",
    color = "Log_Preis",
    size = "Reviews"
  ) +
  theme_minimal()
```



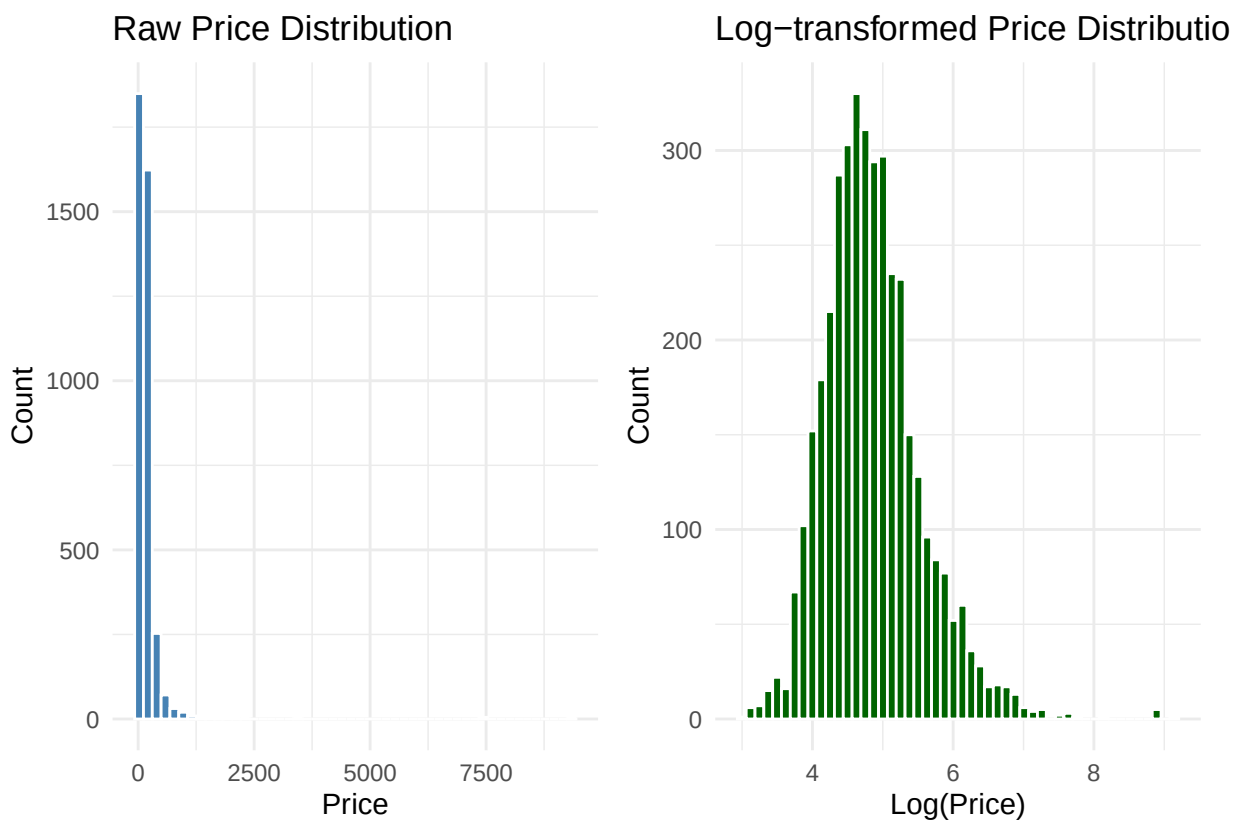
Important to mention During our exploratory data analysis, we identified one 'Private room' listing with an unrealistically high price of 2'147'483'647 CHF. We excluded this single observation from our dataset as it is an extreme outlier that does not reflect the true market conditions and would negatively skew our predictive models.

```
# This keeps everything EXCEPT the row that is a Private room AND costs more than 5000
d.airbnb_listings_feat <- d.airbnb_listings_feat %>%
  filter(!(room_type == "Private room" & price > 100000))
```

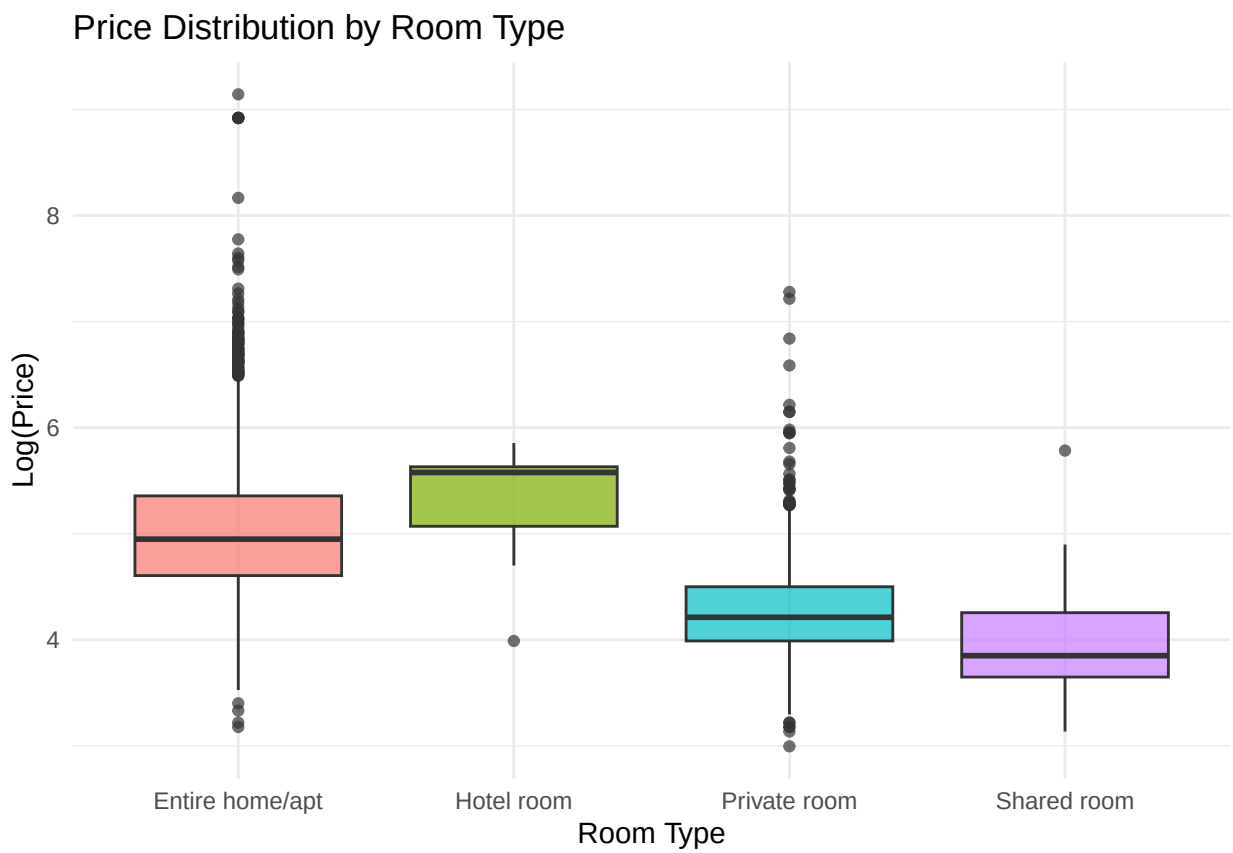
Davide's EDA

```
# --- 1. Distribution of the Target Variable (Price) ---
# Upgrading base R hist() to ggplot2 for a more professional look.
# Showing both raw and log side-by-side helps visually justify the transformation to the client.
p_raw <- ggplot(d.airbnb_listings_feat, aes(x = price)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Raw Price Distribution", x = "Price", y = "Count") +
  theme_minimal()

p_log <- ggplot(d.airbnb_listings_feat, aes(x = log_price)) +
  geom_histogram(bins = 50, fill = "darkgreen", color = "white") +
  labs(title = "Log-transformed Price Distribution", x = "Log(Price)", y = "Count") +
  theme_minimal()
p_raw + p_log # Displays them side-by-side
```

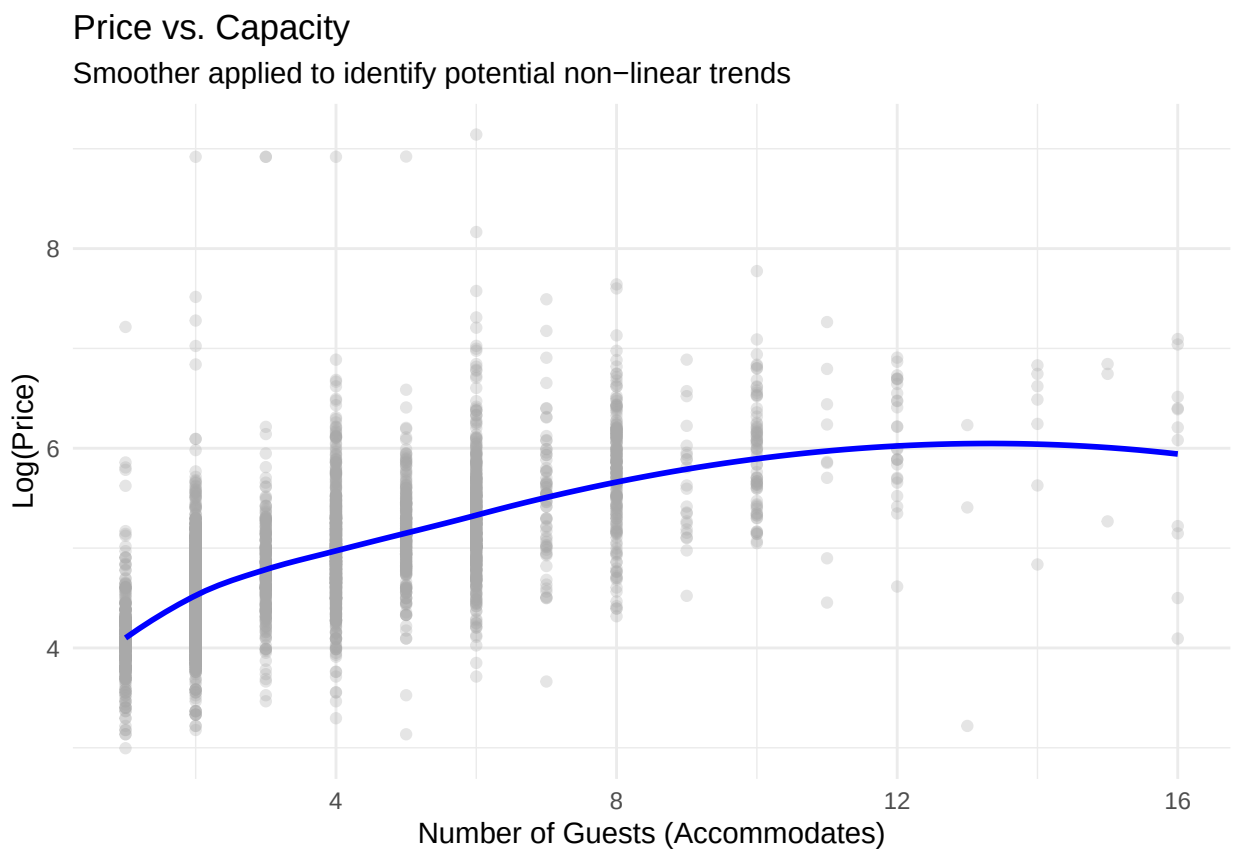



```r
# --- 2. Categorical Predictor: Room Type vs. Price ---
# Your original code was good, just cleaned up the aesthetics.
ggplot(data = d.airbnb_listings_feat, aes(x = room_type, y = log_price, fill = room_type)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    title = "Price Distribution by Room Type",
    x = "Room Type",
    y = "Log(Price)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



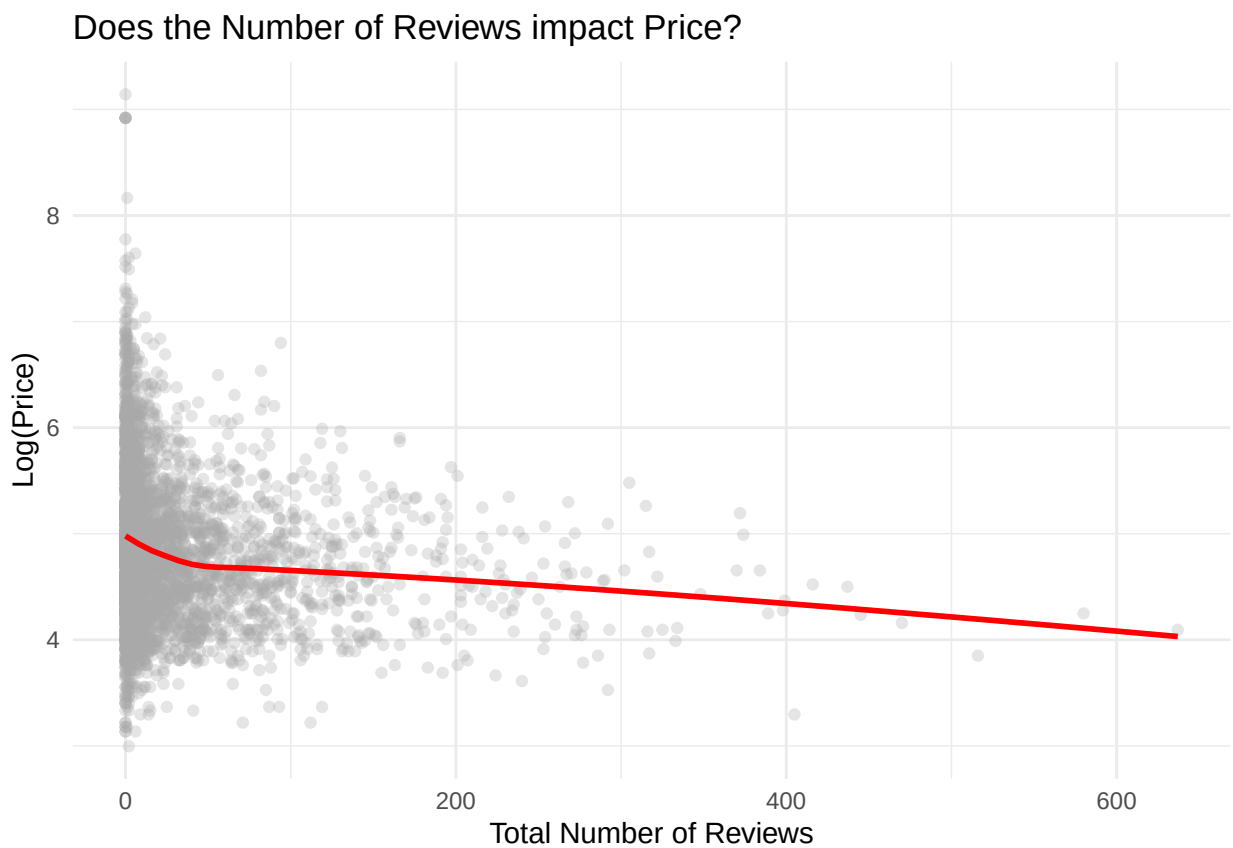

```r
# --- 3. Continuous Predictor: Capacity vs. Price ---
ggplot(data = d.airbnb_listings_feat, aes(x = accommodates, y = log_price)) +
  geom_point(alpha = 0.3, color = "darkgray") +
  geom_smooth(method = "loess", color = "blue", se = FALSE) +
  labs(
    title = "Price vs. Capacity",
    subtitle = "Smoother applied to identify potential non-linear trends",
    x = "Number of Guests (Accommodates)",
    y = "Log(Price)"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

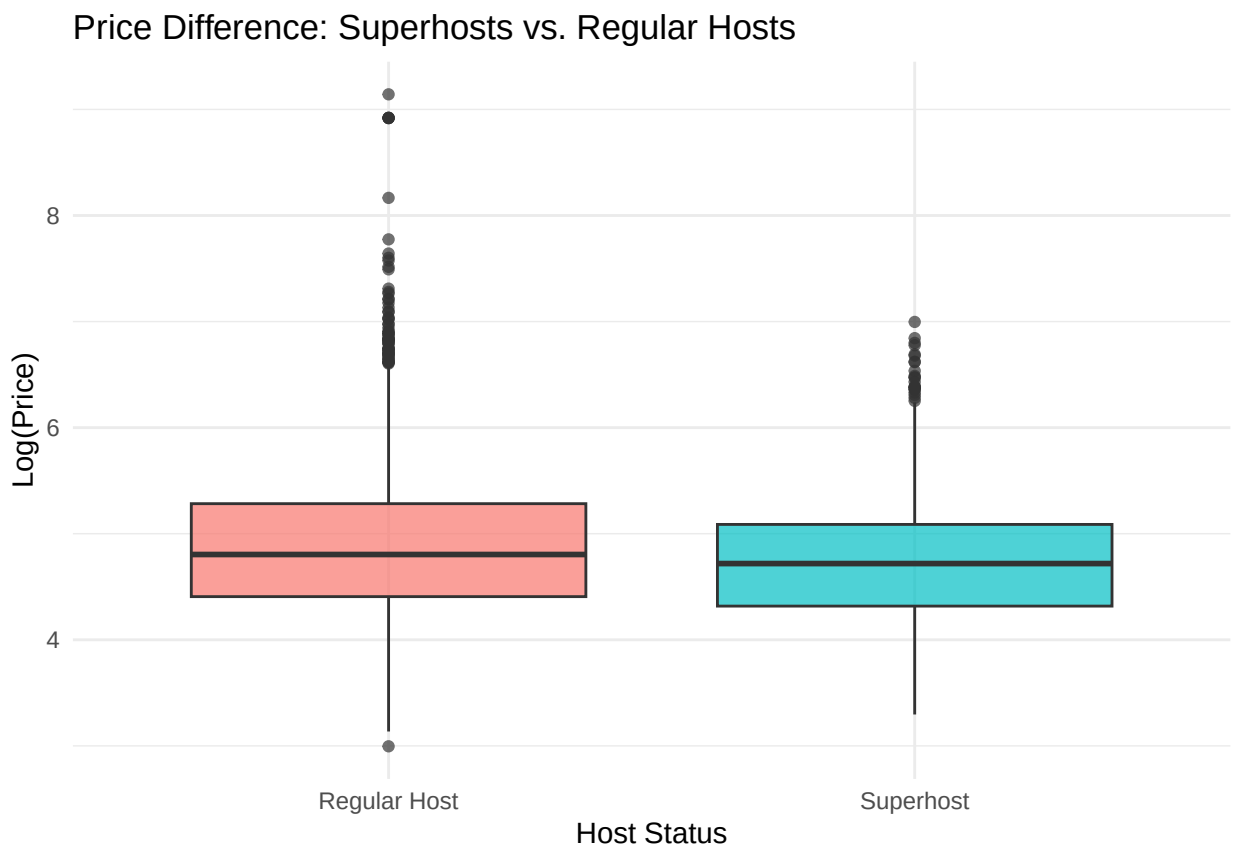



```r
# --- 4. Continuous Predictor: Reviews vs. Price ---
# Correction: The response variable (log_price) should generally be on the y-axis
# when plotting against a predictor[cite: 255].
ggplot(data = d.airbnb_listings_feat, aes(x = number_of_reviews, y = log_price)) +
  geom_point(alpha = 0.3, color = "darkgray") +
  geom_smooth(method = "loess", color = "red", se = FALSE) +
  labs(
    title = "Does the Number of Reviews impact Price?",
    x = "Total Number of Reviews",
    y = "Log(Price)"
  ) +
  theme_minimal()
```

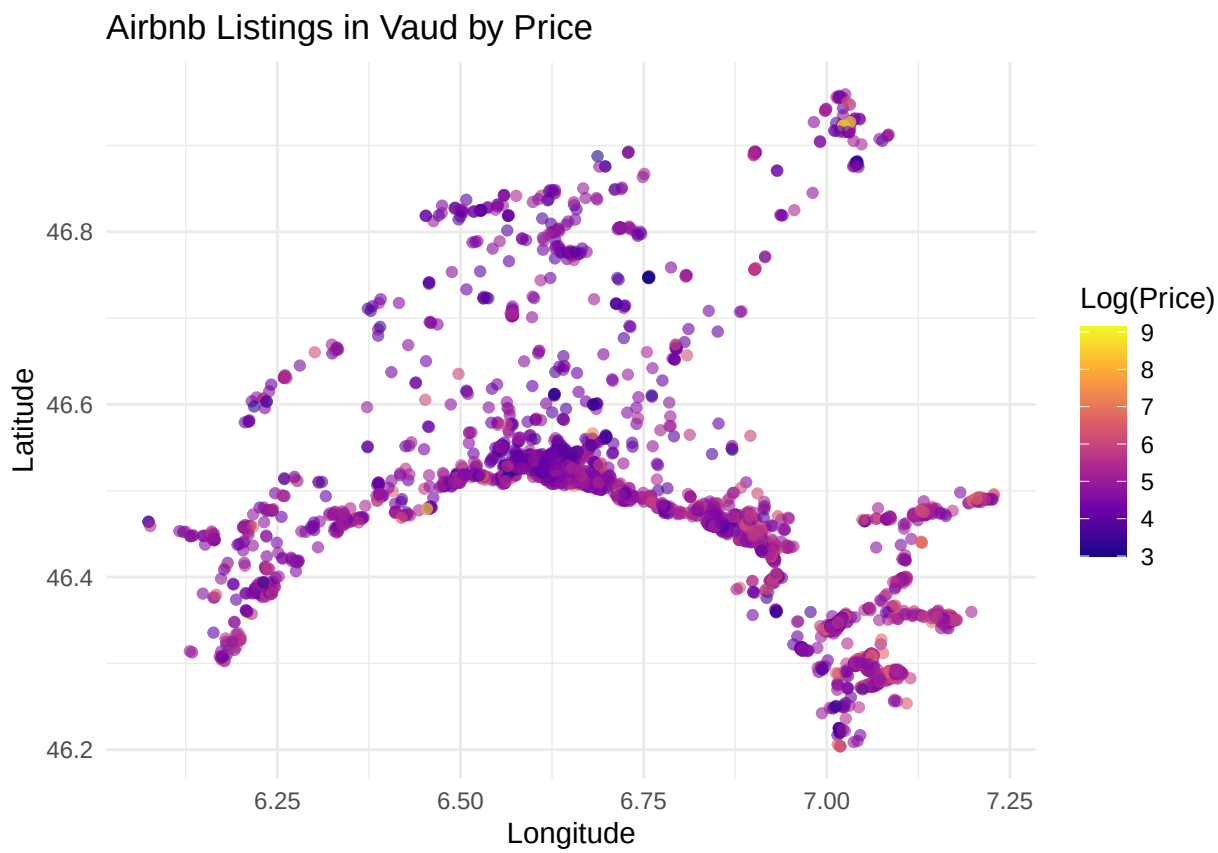
```
`geom_smooth()` using formula = 'y ~ x'
```




```
# --- 5. Binary Predictor: Superhost Status ---
# Adding this because it's a key part of your business case and fulfills the binary variable requiremen
ggplot(data = d.airbnb_listings_feat, aes(x = as.factor(host_is_superhost), y = log_price, fill = as.fa
  geom_boxplot(alpha = 0.7) +
  scale_x_discrete(labels = c("0" = "Regular Host", "1" = "Superhost")) +
  labs(
    title = "Price Difference: Superhosts vs. Regular Hosts",
    x = "Host Status",
    y = "Log(Price)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```




```r
# --- 6. Spatial Data: Mapping the Listings ---
# Cleaned up the map to focus purely on location and price.
ggplot(data = d.airbnb_listings_feat, aes(x = longitude, y = latitude)) +
  geom_point(aes(color = log_price), alpha = 0.6, size = 1.5) +
  scale_color_viridis_c(option = "plasma") +
  labs(
    title = "Airbnb Listings in Vaud by Price",
    x = "Longitude",
    y = "Latitude",
    color = "Log(Price)"
  ) +
  theme_minimal()
```



Davide's overall Data Analysis ## 1. Overall Market Overview (Prices & Capacity) This is the absolute minimum, maximum, median, and mean for the most important numeric variables.

```
# Basic summary statistics for key numerical variables
summary_stats <- d.airbnb_listings_feat %>%
  select(price, log_price, accommodates, number_of_reviews) %>%
  summary()

print(summary_stats)
```

```
     price          log_price      accommodates    number_of_reviews
 Min.   :  20.0   Min.   :2.996   Min.   : 1.000   Min.   :  0.00
 1st Qu.:  80.0   1st Qu.:4.382   1st Qu.: 2.000   1st Qu.:  1.00
 Median : 120.0   Median :4.787   Median : 3.000   Median :  7.00
 Mean   : 178.7   Mean   :4.858   Mean   : 3.755   Mean   : 29.52
 3rd Qu.: 185.0   3rd Qu.:5.220   3rd Qu.: 5.000   3rd Qu.: 32.00
 Max.   :9342.0   Max.   :9.142   Max.   :16.000   Max.   :637.00
```

Client Interpretation: "The average Airbnb in Vaud costs X CHF per night, but the median is Y CHF, indicating that a few extremely expensive luxury listings might be skewing the average upward. Prices range from a minimum of A CHF to a maximum of B CHF."

2. Analysis by Room Type

Since we want to know how many different types of rooms there are, we group the data and calculate the counts, percentages, and price ranges for each type.

```r
# Detailed breakdown by Room Type
room_type_analysis <- d.airbnb_listings_feat %>%
  group_by(room_type) %>%
  summarise(
    Total_Listings = n(),
    Percentage_of_Market = round((n() / nrow(d.airbnb_listings_feat)) * 100, 1),
    Min_Price = min(price, na.rm = TRUE),
    Median_Price = median(price, na.rm = TRUE),
    Average_Price = round(mean(price, na.rm = TRUE), 1),
    Max_Price = max(price, na.rm = TRUE)
  ) %>%
  arrange(desc(Total_Listings)) # Sorts by the most common room type

print(room_type_analysis)
```

```
# A tibble: 4 x 7
  room_type       Total_Listings Percentage_of_Market Min_Price Median_Price
  <fct>                    <int>                <dbl>     <dbl>        <dbl>
1 Entire home/apt           2908                 75          24        141
2 Private room               940                 24.3        20         67.5
3 Shared room                 19                  0.5        23         47
4 Hotel room                   8                  0.2        54        264.
# i 2 more variables: Average_Price <dbl>, Max_Price <dbl>
```

Client Interpretation: "Entire homes/apartments dominate the Vaud market, making up 75% of all listings. As expected, the Hotel room command the highest median price (264.5 CHF), compared to shared rooms (47 CHF)."

3. Superhost vs. Regular Host Breakdown

This business case specifically mentions superhost status. So we took a look to see how many there are and if they actually charge more or even less.

```r
# Superhost statistics
superhost_analysis <- d.airbnb_listings_feat %>%
  group_by(host_is_superhost) %>%
  summarise(
    Total_Hosts = n(),
    Percentage = round((n() / nrow(d.airbnb_listings_feat)) * 100, 1),
    Median_Price = median(price, na.rm = TRUE),
    Avg_Reviews = round(mean(number_of_reviews, na.rm = TRUE), 1)
  ) %>%
  mutate(host_is_superhost = ifelse(host_is_superhost == 1, "Yes", "No"))

print(superhost_analysis)
```

```
# A tibble: 2 x 5
  host_is_superhost Total_Hosts Percentage Median_Price Avg_Reviews
  <chr>                   <int>      <dbl>        <dbl>       <dbl>
1 No                       2806       72.4          122        17.3
2 Yes                      1069       27.6          112        61.6
```

---

das kann glaub gelöscht werden - Davide!!!!*****

1. Host Experience vs. Price (host_since_year)

Now let's test the hypothesis: can "older" (more experienced) hosts charge higher prices than newcomers?

```
# First, ensure the year is treated as a numeric variable
#d.airbnb_listings_feat <- d.airbnb_listings_feat %>%
#  mutate(host_since_year = as.numeric(host_since_year))

# Plot: Price trend over the years of host membership
#ggplot(data = d.airbnb_listings_feat %>% filter(!is.na(host_since_year)),
#       aes(x = host_since_year, y = log_price)) +
#  geom_point(alpha = 0.2, color = "darkcyan") +
  # Adding a smoother is mandatory for exploratory plots according to the hints!
#  geom_smooth(method = "gam", formula = y ~ s(x, bs = "cs"), color = "orange", se = TRUE) +
#  labs(
#    title = "Can established hosts command higher prices?",
#    x = "Host Since (Year)",
#    y = "Log(Price)"
#  ) +
#  theme_minimal()
```

Client Interpretation: "We observe that hosts who joined the platform earlier (e.g., around 2015) do (or do not) command higher average prices compared to recent joiners. For our pricing tool, this means…"

2. Correlation Matrix (Crucial before Machine Learning)

Before we train our LM, GAM, or SVM, we need to know how our numerical variables relate to each other. This helps us identify multicollinearity (when two predictors provide the same information) and strong price drivers.

```
# Select relevant numerical variables (drop NAs just for the correlation calculation)
cor_data <- d.airbnb_listings_feat %>%
  select(log_price, accommodates, number_of_reviews) %>%
  drop_na()

# Calculate the correlation matrix
cor_matrix <- cor(cor_data)

# Render a nice table for your RMarkdown document
kable(round(cor_matrix, 2), caption = "Correlation Matrix of Key Numerical Drivers")
```

Table 1: Correlation Matrix of Key Numerical Drivers

	log_price	accommodates	number_of_reviews
log_price	1.00	0.65	-0.16
accommodates	0.65	1.00	-0.04
number_of_reviews	-0.16	-0.04	1.00

Client Interpretation: "As expected, accommodates shows the strongest positive correlation with price. It is also interesting to note the relationship between availability and number_of_reviews…"

3. Detailed property_type Analysis

We already analyzed room_type, but Airbnb often has dozens of property_types (e.g., "Entire chalet", "Private room in castle", etc.). Summarizing the Top 10 property types in the Canton of Vaud provides

maybe a great market insight.

```
# Top 10 Property Types by frequency and their median price
top_properties <- d.airbnb_listings_feat %>%
  group_by(property_type) %>%
  summarise(
    Total_Listings = n(),
    Median_Price = median(price, na.rm = TRUE)
  ) %>%
  arrange(desc(Total_Listings)) %>%
  slice_head(n = 10) # Keep only the top 10

kable(top_properties, caption = "Top 10 Property Types in the Canton of Vaud")
```

Table 2: Top 10 Property Types in the Canton of Vaud

property_type	Total_Listings	Median_Price
Entire rental unit	1974	128.0
Private room in rental unit	409	64.0
Entire home	287	250.0
Entire chalet	219	250.0
Private room in bed and breakfast	194	69.5
Entire condo	184	136.5
Private room in home	154	66.0
Entire villa	50	370.0
Private room in villa	45	68.0
Room in hotel	40	177.0

Client Interpretation: "While standard apartments like"entire rental unit" are the most common, specialty properties like chalets, homes and villas command a significant premium, which our model will need to capture."

4. Missing Values (NA) Check

This step is less for the client and essential for your ML models. Most modeling functions (like lm() or svm()) will throw errors or silently drop entire rows if NAs are present. You need to know exactly where your data is missing!

```
# Count the number of NAs for each key column in your dataset
na_summary <- d.airbnb_listings_feat %>%
  select(price, log_price, accommodates, number_of_reviews, host_is_superhost) %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  t() %>% # Transpose for better readability
  as.data.frame() %>%
  rename(Missing_Values = V1)

kable(na_summary, caption = "Count of Missing Values per Variable")
```

Table 3: Count of Missing Values per Variable

	Missing_Values
price	0
log_price	0

	Missing_Values
accommodates	0
number_of_reviews	0
host_is_superhost	0

##AI-Generated Models by Davide Bi gad Moti gsi Davide: Linear Model Before we jump straight into the lm() function, we need to do one crucial step for your business case: Train/Test Splitting. Since your goal is to build a predictive pricing tool, you need to train your model on one chunk of the data (e.g., 80%) and test its accuracy on data it has never seen before (the remaining 20%). This proves to the client that your tool actually works in the real world.Additionally, the evaluation guidelines explicitly warn against fitting a "very simple linear model" if you are going to compare it to complex models like Neural Networks or SVMs. To make it a "fair way" of comparing, you should include at least one interaction term in your LM.

```
# --- 1. Split Data into Training and Testing Sets ---
# Using the caret package (which you already loaded) to create an 80/20 split
#set.seed(42) # Set a seed so you get the exact same split every time you compile!
#train_index <- createDataPartition(d.airbnb_listings_feat$log_price, p = 0.8, list = FALSE)

#train_data <- d.airbnb_listings_feat[train_index, ]
#test_data <- d.airbnb_listings_feat[-train_index, ]

# --- 2. Fit the Linear Model ---
# Note: We MUST use log_price because "amounts" must virtually always be log-transformed[cite: 192, 193
# We add an interaction term (*) between accommodates and room_type for a fairer comparison with comple
#lm_model <- lm(log_price ~ accommodates * room_type +
#                 host_is_superhost +
#                 number_of_reviews +
#                 availability_365,
#               data = train_data)

# --- 3. View the Results ---
#summary(lm_model)
```

Interpreting the Coefficients

Because your target variable is log_price, interpreting the coefficients requires a slight mental shift. A coefficient in a log-linear model represents an *approximate percentage change in the original price.*

The Baseline (The Intercept)

What the math says: The intercept is 4.371. *Client Translation:* Our baseline reference listing—an 'Entire home/apt' (the category missing from the list, meaning it's the default)—has an estimated base price of roughly 79 CHF per night (exp(4.371)), assuming all other factors are zero.

Capacity (Accommodates)

What the math says: The coefficient is 0.1553 (Highly significant). *Client Translation:* Capacity is a massive price driver. For a standard entire apartment, every additional guest the property can accommodate increases the nightly price by 16.8% ((exp(0.1553) - 1) * 100).

Room Types & The Interaction Term

This is where your interaction term (accommodates * room_type) pays off, proving why you shouldn't use a simple, plain model!

Private Rooms: They are inherently cheaper. The base price drops by roughly 30.8% (exp(-0.3688) - 1) compared to an entire home. The interaction term for private rooms is not statistically significant (p = 0.52), meaning an extra guest in a private room adds about the same 16.8% premium as it does in an entire home.

Shared Rooms: This is the most interesting part. The base price drops, but look at the interaction term: -0.1585 (Highly significant). If you add the base guest coefficient (0.1553) to this shared room interaction (-0.1585), you get essentially zero.

Client Translation: "While adding guests allows hosts of 'Entire homes' and 'Private rooms' to charge about 17% more per guest, this rule does not apply to 'Shared rooms'. For shared rooms, adding more beds does not allow the host to increase the price at all.

Surprising Business Insights (Other Drivers)

Superhost Status (-0.0648): "Counter-intuitively, Superhosts actually charge about 6.3% less than regular hosts. From a business perspective, this suggests that hosts might achieve Superhost status because they offer highly competitive, slightly lower prices to ensure high booking rates and good reviews."

Number of Reviews (-0.0014): "Listings with more reviews tend to be slightly cheaper (each review drops the price by about 0.14%). This likely points to 'budget' listings having a higher turnover of guests, and therefore accumulating more reviews."

Availability (0.00029): "Listings with higher availability are slightly more expensive. This makes sense: highly expensive places are booked less often, leaving more days available on their calendar."

Evaluating Predictive Performance

To see how "accurate" our model is, we will calculate the Root Mean Squared Error (RMSE) on the test data. The guidelines state that if we cannot find another measure of fit, we can use RMSE.

```
# --- 4. Make Predictions on the Test Data ---
# Predict the log_price for the unseen test data
#lm_predictions <- predict(lm_model, newdata = test_data)

# FIX: Force R to use the Metrics package specifically
#lm_rmse <- Metrics::rmse(test_data$log_price, lm_predictions)

#print(paste("Linear Model RMSE on Test Data:", round(lm_rmse, 4)))
```

Client Translation: "Our baseline Linear Model achieved a Root Mean Squared Error (RMSE) of 0.4624 on unseen test data. Because we are predicting the logarithm of the price, this means our model's predictions are typically off by a factor of roughly exp(0.4624), which equals about 1.58. In practical terms, our base model's price estimates are generally within a ~58% margin of error. While this is a good starting point, it highlights the need for our more advanced machine learning models to capture the complex, non-linear pricing dynamics in the Vaud market."

#LM (Timon)

```
#das lm erstellen.

lm.overall <- lm(price ~ as.factor(host_response_time) + host_response_rate + host_acceptance_rate + as
                                  as.factor(host_has_profile_pic) + as.factor(host_identity_verified) +
                                  as.factor(room_type) + bathrooms + beds, minimum_nights + number_of_
                                  review_scores_rating, data = d.airbnb_listings_feat)
summary(lm.overall)
```

```
Call:
lm(formula = price ~ as.factor(host_response_time) + host_response_rate +
```

```
    host_acceptance_rate + as.factor(host_is_superhost) + as.factor(host_has_profile_pic) +
    as.factor(host_identity_verified) + as.factor(property_type) +
    as.factor(room_type) + bathrooms + beds, data = d.airbnb_listings_feat,
    subset = minimum_nights + number_of_reviews + estimated_occupancy_l365d +
        review_scores_rating)

Residuals:
    Min      1Q  Median      3Q     Max
-291.08  -36.00    2.43   22.12 1094.09

Coefficients: (3 not defined because of singularities)
                                                        Estimate Std. Error
(Intercept)                                              78.2889   116.3981
as.factor(host_response_time)within a day               -22.4261    19.3396
as.factor(host_response_time)within a few hours         -43.9188    20.2223
as.factor(host_response_time)within an hour             -40.7238    20.9740
host_response_rate                                      120.7929    20.4073
host_acceptance_rate                                     -9.2186     8.9109
as.factor(host_is_superhost)1                           -27.6970     4.0761
as.factor(host_has_profile_pic)t                         11.6875    84.3637
as.factor(host_identity_verified)t                      -60.7784    16.9460
as.factor(property_type)Entire chalet                   -23.6185    82.3929
as.factor(property_type)Entire condo                    -32.8345    82.5084
as.factor(property_type)Entire guesthouse               -35.7281    83.3311
as.factor(property_type)Entire home                      26.9555    82.5592
as.factor(property_type)Entire loft                     -46.7404    84.7066
as.factor(property_type)Entire place                    -95.0824   100.8745
as.factor(property_type)Entire rental unit              -47.1387    82.2649
as.factor(property_type)Entire serviced apartment       -27.5664    83.7721
as.factor(property_type)Entire townhouse               -109.7596    84.1591
as.factor(property_type)Entire villa                    -28.4894    84.8883
as.factor(property_type)Private room in bed and breakfast  -82.4092    82.5254
as.factor(property_type)Private room in chalet         -143.9007    85.9819
as.factor(property_type)Private room in condo          -114.5967    92.0031
as.factor(property_type)Private room in cottage        -121.3965   116.2728
as.factor(property_type)Private room in guesthouse     -158.9179   100.6608
as.factor(property_type)Private room in home            -85.1213    82.5087
as.factor(property_type)Private room in loft             11.9181    95.0242
as.factor(property_type)Private room in rental unit    -112.3338    82.3647
as.factor(property_type)Private room in townhouse      -117.8635    85.9907
as.factor(property_type)Private room in villa          -134.3734    82.5055
as.factor(property_type)Room in bed and breakfast        83.0774    85.3332
as.factor(property_type)Room in hotel                   -58.3965    90.1033
as.factor(property_type)Shared room in bed and breakfast  -208.7675    90.4155
as.factor(room_type)Hotel room                                 NA         NA
as.factor(room_type)Private room                               NA         NA
as.factor(room_type)Shared room                                NA         NA
bathrooms                                                92.5146     3.4576
beds                                                     -1.4697     0.6375
                                                       t value Pr(>|t|)
(Intercept)                                              0.673 0.501262
as.factor(host_response_time)within a day               -1.160 0.246316
as.factor(host_response_time)within a few hours         -2.172 0.029958 *
as.factor(host_response_time)within an hour             -1.942 0.052286 .
```

```
host_response_rate                                          5.919 3.65e-09 ***
host_acceptance_rate                                       -1.035 0.300978
as.factor(host_is_superhost)1                              -6.795 1.33e-11 ***
as.factor(host_has_profile_pic)t                            0.139 0.889826
as.factor(host_identity_verified)t                         -3.587 0.000341 ***
as.factor(property_type)Entire chalet                      -0.287 0.774397
as.factor(property_type)Entire condo                       -0.398 0.690696
as.factor(property_type)Entire guesthouse                  -0.429 0.668140
as.factor(property_type)Entire home                         0.326 0.744072
as.factor(property_type)Entire loft                        -0.552 0.581137
as.factor(property_type)Entire place                       -0.943 0.345979
as.factor(property_type)Entire rental unit                 -0.573 0.566685
as.factor(property_type)Entire serviced apartment          -0.329 0.742133
as.factor(property_type)Entire townhouse                   -1.304 0.192279
as.factor(property_type)Entire villa                       -0.336 0.737191
as.factor(property_type)Private room in bed and breakfast  -0.999 0.318082
as.factor(property_type)Private room in chalet             -1.674 0.094322 .
as.factor(property_type)Private room in condo              -1.246 0.213029
as.factor(property_type)Private room in cottage            -1.044 0.296548
as.factor(property_type)Private room in guesthouse         -1.579 0.114511
as.factor(property_type)Private room in home               -1.032 0.302322
as.factor(property_type)Private room in loft                0.125 0.900199
as.factor(property_type)Private room in rental unit        -1.364 0.172726
as.factor(property_type)Private room in townhouse          -1.371 0.170597
as.factor(property_type)Private room in villa              -1.629 0.103502
as.factor(property_type)Room in bed and breakfast           0.974 0.330360
as.factor(property_type)Room in hotel                      -0.648 0.516972
as.factor(property_type)Shared room in bed and breakfast   -2.309 0.021020 *
as.factor(room_type)Hotel room                                 NA       NA
as.factor(room_type)Private room                               NA       NA
as.factor(room_type)Shared room                                NA       NA
bathrooms                                                  26.757  < 2e-16 ***
beds                                                       -2.306 0.021213 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 82.16 on 2701 degrees of freedom
  (1140 Beobachtungen als fehlend gelöscht)
Multiple R-squared:  0.5534,    Adjusted R-squared:  0.5479
F-statistic: 101.4 on 33 and 2701 DF,  p-value: < 2.2e-16
```

```r
#fehlt: host_verifications, neighbourhood_group_cleansed, latitde, longitude, accomodates, log_price
#Diese Variablen wurden aus verschiedenen Gründen nicht miteingebaut.
```

Vorläufiges Fazit: - die Variablen müssen noch angepasst werden. beispielsweise macht es möglicherweise sinn, private Rooms zusammenzufassen. ist ja eigentlich egal in welcher ARt gebäude. - wieso hat room_type so viele NA?

GLM set to poisson (Pascal)

```r
# Poisson-Modell zum Auslastung vorherzusagen

glm_poisson <- glm(estimated_occupancy_l365d ~ price + host_is_superhost +
                     review_scores_rating + accommodates + room_type,
```

```r
                    data = d.airbnb_listings_feat,
                    family = "poisson")

summary(glm_poisson)
```

```
Call:
glm(formula = estimated_occupancy_l365d ~ price + host_is_superhost +
    review_scores_rating + accommodates + room_type, family = "poisson",
    data = d.airbnb_listings_feat)

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)            2.806e+00  4.323e-02  64.908  < 2e-16 ***
price                 -3.064e-03  3.744e-05 -81.836  < 2e-16 ***
host_is_superhost      9.577e-01  5.062e-03 189.212  < 2e-16 ***
review_scores_rating   2.256e-01  9.140e-03  24.680  < 2e-16 ***
accommodates           3.845e-02  1.433e-03  26.837  < 2e-16 ***
room_typeHotel room    5.530e-01  3.811e-02  14.509  < 2e-16 ***
room_typePrivate room -1.358e-01  6.300e-03 -21.561  < 2e-16 ***
room_typeShared room  -1.880e-01  3.530e-02  -5.326 1.01e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 233523  on 3160  degrees of freedom
Residual deviance: 178223  on 3153  degrees of freedom
  (714 Beobachtungen als fehlend gelöscht)
AIC: 192251

Number of Fisher Scoring iterations: 6
```

```r
# Gibt es Overdispersion? Poisson-Annahme ist, dass Mittelwert exakt gleich der Varianz ist. Airbnbs si
dispersion_ratio <- glm_poisson$deviance / glm_poisson$df.residual
print(paste("Dispersion Ratio:", dispersion_ratio))
```

```
[1] "Dispersion Ratio: 56.525007402987"
```

```r
glm_quasipoisson <- glm(estimated_occupancy_l365d ~ price + host_is_superhost +
                          review_scores_rating + accommodates + room_type,
                        data = d.airbnb_listings_feat,
                        family = "quasipoisson")

summary(glm_quasipoisson)
```

```
Call:
glm(formula = estimated_occupancy_l365d ~ price + host_is_superhost +
    review_scores_rating + accommodates + room_type, family = "quasipoisson",
    data = d.airbnb_listings_feat)

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            2.8062737  0.3666905   7.653 2.59e-14 ***
```

```
price                 -0.0030635  0.0003175  -9.649  < 2e-16 ***
host_is_superhost      0.9577253  0.0429297  22.309  < 2e-16 ***
review_scores_rating   0.2255679  0.0775179   2.910  0.00364 **
accommodates           0.0384455  0.0121500   3.164  0.00157 **
room_typeHotel room    0.5530131  0.3232619   1.711  0.08723 .
room_typePrivate room -0.1358318  0.0534313  -2.542  0.01106 *
room_typeShared room  -0.1879854  0.2993700  -0.628  0.53009
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 71.93338)

    Null deviance: 233523  on 3160  degrees of freedom
Residual deviance: 178223  on 3153  degrees of freedom
  (714 Beobachtungen als fehlend gelöscht)
AIC: NA

Number of Fisher Scoring iterations: 6
```

Interpretation

GLM-Poisson:

Das GLM mit Poissonverteilung hat wie erwartet eine Überdispersion produziert, da die Streuung der Airbnb-Daten zu erwarten war. Das Modell hat die Standardfehler zu klein berechnet, wweshalb im ersten Output alle Variablen hochsignifikant waren. Danach wurde die Dispersion berechnet und sie war grösser als 1, nämlich 56.53. Als Lösung wurde ein Quasipoissonmodell genutzt, welches die Standardfehler nach oben anpasst.

GLM-Quasipoisson

Das GLM mit Quasipoissonverteilung hat die Überdispersion korrigiert. Die Standardfehler wurden nach oben angepasst (deutlich) und gibt nun ein realistisches Bild, welche Faktoren wirklich einen Einfluss auf die Auslastung haben. Interpretationen: - Price -> hochsignifikant negativ. Wie erwartet: Je höher der Preis, desto geringer die Auslastung - Superhost: hochsignifikant positiv: Achtung, logarithmierte Raten, darum Effekt mit Exponentialfunktion berechnen $\exp(0.9577) = 2.6$. Ein Superhost zu sein steigert die Auslastung um das 2,6-fache im Vergleich zum normalen Host. - Bewertungen: Positiv und signifikant. Bessere Bewertungen führen zu mehr Buchungen. - Kapazität: Positiv und signifikant. Je mehr Leute reinpassen, desto mehr Buchungen. - Zimmertypen: Im Intercept ist "Entire home/apt". Private Room ist signifikant schlechter gebucht als ganze Unterkunft. Shared Room hat hohen P-Wert. Statistisch kann man nicht sagen, ob sich die Auslastung eines geteilten Zimmers von der ganzen Wochnung unterscheidet.

GLM Quasi-Poisson Ergebnisse für Businesscase

```r
# Extract model results and calculate the percentage change for each variable
d.results_glm_quasipoisson <- tidy(glm_quasipoisson) %>%
  # Calculate the percentage
  mutate(
    pct_change = (exp(estimate) - 1) * 100,
    # Round the percentage change to 2
    pct_change = round(pct_change, 2)
  ) %>%
  # Select only the relevant columns for business case
  select(term, estimate, pct_change, p.value)
```

```r
print(d.results_glm_quasipoisson)
```

```
# A tibble: 8 x 4
  term                  estimate pct_change   p.value
  <chr>                    <dbl>      <dbl>     <dbl>
1 (Intercept)            2.81       1555.   2.59e- 14
2 price                 -0.00306      -0.31 9.89e- 22
3 host_is_superhost      0.958       161.   1.73e-102
4 review_scores_rating   0.226        25.3  3.64e-  3
5 accommodates           0.0384        3.92 1.57e-  3
6 room_typeHotel room    0.553        73.8  8.72e-  2
7 room_typePrivate room -0.136       -12.7  1.11e-  2
8 room_typeShared room  -0.188       -17.1  5.30e-  1
```

#GLM set to binomial(Davide) 1. Schritt: Zielvariable binarisieren Aus Preis eine 0/1-Variable machen Der Median ist der sauberste Cutpoint — er ist robust gegenüber Ausreissern und teilt den Datensatz exakt in zwei Hälften

```r
d.airbnb_listings_feat <- d.airbnb_listings_feat %>%
  mutate(price_high = as.integer(price > median(price, na.rm = TRUE)))

# Checken ob Split sauber ist
table(d.airbnb_listings_feat$price_high)
```

```
   0    1
1980 1895
```

2. Schritt GLM Binomial

```r
glm_binomial <- glm(price_high ~ host_is_superhost,
                    data   = d.airbnb_listings_feat,
                    family = binomial(link = "logit"))

summary(glm_binomial)
```

```
Call:
glm(formula = price_high ~ host_is_superhost, family = binomial(link = "logit"),
    data = d.airbnb_listings_feat)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        0.01711    0.03776   0.453  0.65050
host_is_superhost -0.22175    0.07216  -3.073  0.00212 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5370.0  on 3874  degrees of freedom
Residual deviance: 5360.6  on 3873  degrees of freedom
AIC: 5364.6

Number of Fisher Scoring iterations: 3
```

3. Schritt Interpretation - Odds Ratio berechnen Genau wie Pascal exp() für seinen Quasipoisson gebraucht hat, brauchst du das hier auch. Bei Binomial sind es Odds Ratios:

```
d.results_glm_binomial <- tidy(glm_binomial) %>%
  mutate(
    odds_ratio = round(exp(estimate), 3),
    p.value    = round(p.value, 4)
  ) %>%
  select(term, estimate, odds_ratio, p.value)

print(d.results_glm_binomial)
```

```
# A tibble: 2 x 4
  term              estimate odds_ratio p.value
  <chr>                <dbl>      <dbl>   <dbl>
1 (Intercept)         0.0171      1.02   0.650
2 host_is_superhost  -0.222       0.801  0.0021
```

Was der Output sagt Der Koeffizient für host_is_superhost ist negativ (-0.2217) und signifikant (p = 0.002). Das Odds Ratio von 0.801 bedeutet: Superhosts haben ~20% geringere Odds, über dem Medianpreis zu liegen — *somit sind Superhosts tendenziell günstiger, nicht teurer.* Das klingt zuerst wie ein Widerspruch zu H3. Ist es aber nicht. Die eigentlich spannende Erkenntnis: kombiniere das mit Pascals Ergebnis

Pascals Quasipoisson hat gezeigt: Superhost-Status steigert die Auslastung um Faktor 2.6. Dein Binomial zeigt: Superhosts liegen eher unter dem Medianpreis. Das ergibt zusammen ein kohärentes Bild der Superhost-Strategie:

Superhost = niedrigerer Preis + viel höhere Auslastung Normal Host = höherer Preis + deutlich weniger Buchungen

Superhosts optimieren auf Volumen statt Marge. Das macht auch betriebswirtschaftlich Sinn: um den Superhost-Status zu halten, braucht man viele Buchungen und gute Reviews — das funktioniert besser mit kompetitiven Preisen.

H3 wird nicht bestätigt im Sinne von "Superhosts können mehr verlangen". Stattdessen zeigt sich das Gegenteil: *Superhosts preisen strategisch* günstiger, erzielen dafür aber massiv mehr Buchungen (Pascal's Modell). Das ist ein stärkerer Befund als eine simple Bestätigung von H3 — es zeigt eine echte Preisstrategie.

Ein Host der Superhost-Status anstrebt, sollte seinen Preis nicht erhöhen, sondern durch kompetitive Preise Buchungen und Reviews sammeln. Der Superhost-Status zahlt sich dann indirekt über Auslastung aus, nicht über den Einzelpreis.